ENCN442 Surface Water, Groundwater and Coastal Engineering

Surface Water Modelling

Topic of this part of ENCN442

- Weeks 3-6: Hydrological modelling of surface water including hydraulic infrastructure [Assignment released]

Materials for the module on hydrological modelling

Course reader (available on LEARN).

Lecture slides and notes (available on LEARN before/after).

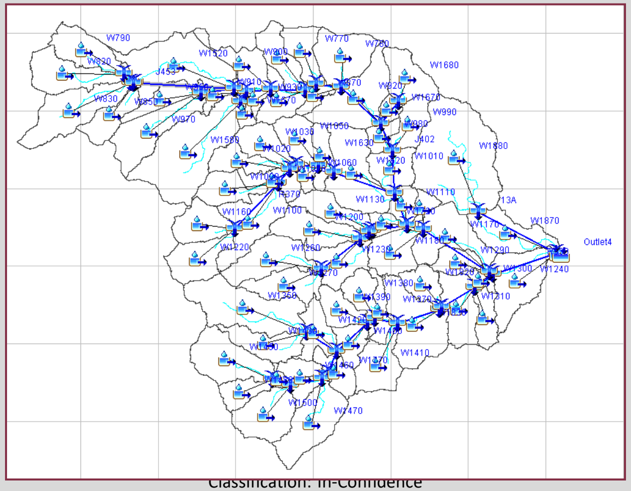
HEC-HMS software and documentation, available at https://www.hec.usace.army.mil/software/hec-hms/ (open access). Installed on computers in various UC computer labs (can be installed via UC 'Software Centre' if not already installed).

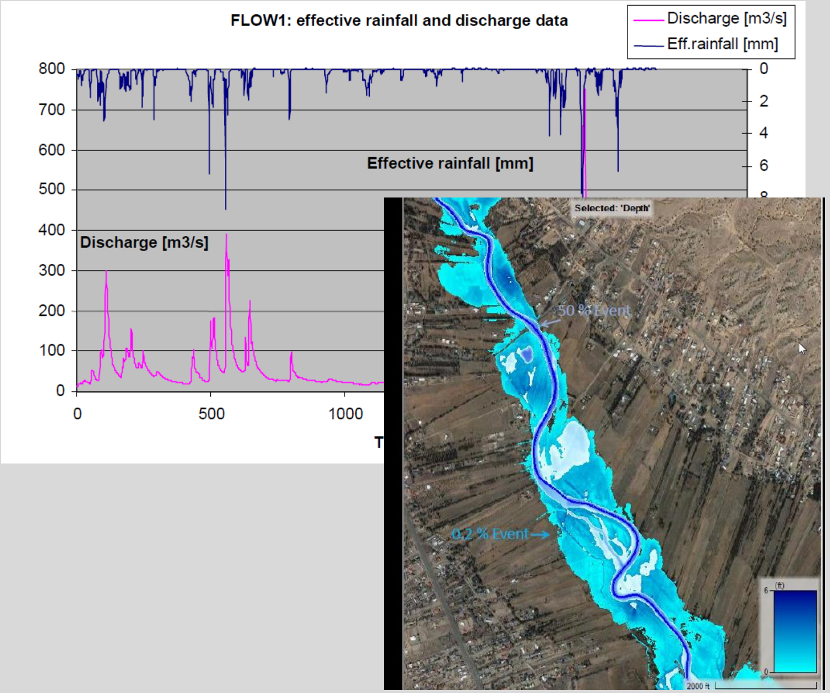
Please also install the software on your own laptops and bring your laptop to the lectures. Microsoft Excel will also be used.

ENCN442 Surface Water, Groundwater and Coastal Engineering

Surface Water Modelling

Hydrological Modelling





Surface Water Modelling

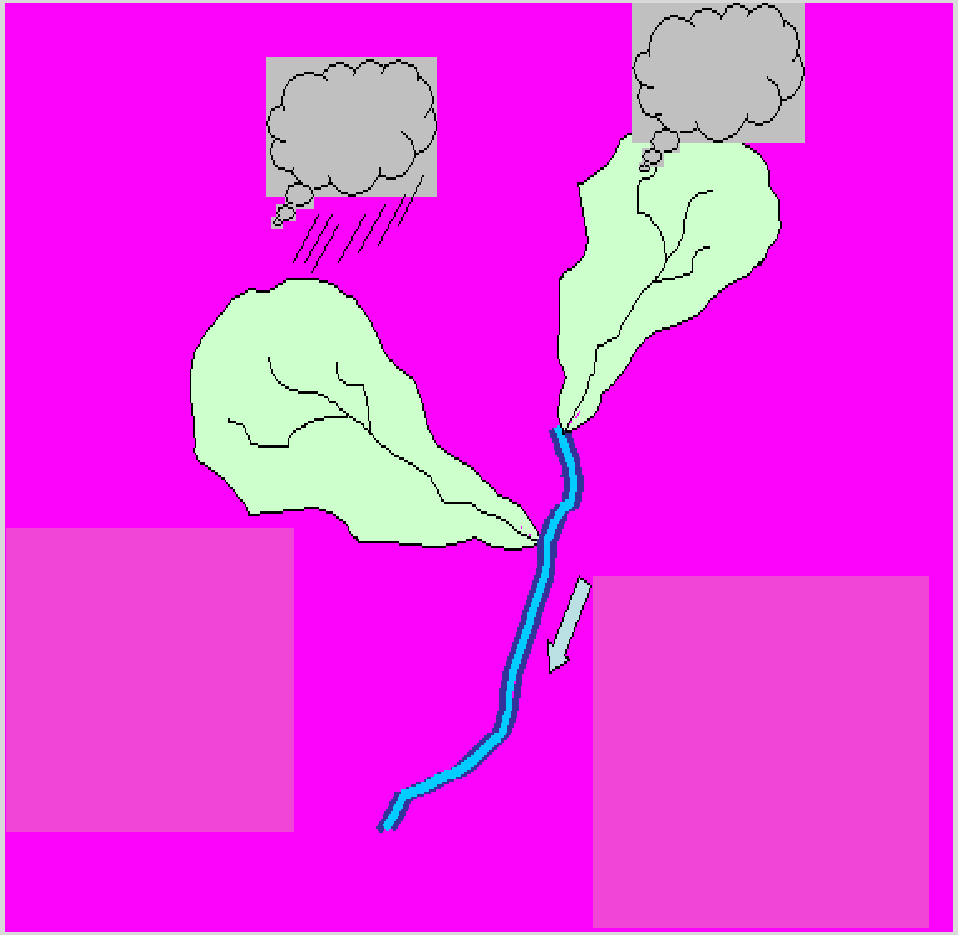
A river system is generally modelled using two types of approaches

- **Hydrological modelling**
- **Hydraulic modelling**

- **Hydrological modelling** is applied in river basins to **transform** precipitation events into discharges in river networks.

- **Hydraulic modelling** involves flow **propagation** in the river, across infrastructure and on floodplains.

- **Combined hydrological and hydraulic modelling:** Blue, green and grey infrastructure (e.g. retention ponds, wetlands, culverts, etc.) is often designed using a combination of hydrological and hydraulic modelling.

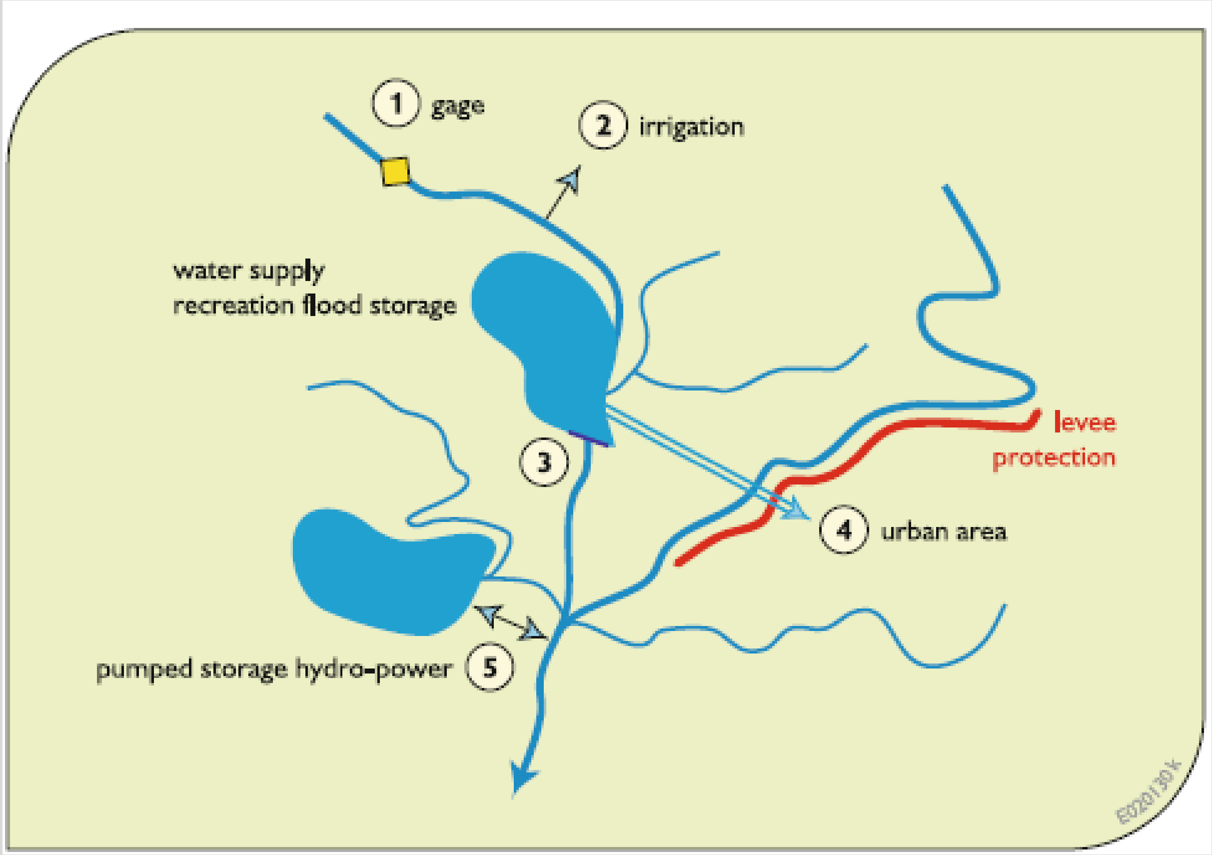


Structure of the module on hydrological modelling

1. **Principles and objectives** of hydrological modelling along with the main **types** of hydrological models and their **structure** and **variables;**
2. **Methods** and **challenges** related to the **representation of a given hydrosystem** and the corresponding hydrological phenomena that occur within it;
3. Problems encountered when **estimating the parameters** of hydrological models;
4. Methods used to **validate models** and characterise the associated **uncertainties;**
5. Choosing a **suitable modelling approach** for a given goal
6. **Development, calibration, validation and application of a hydrological model.**

1.1 Principles and Objectives of Hydrological Modelling

Example: A multipurpose river system



Loucks et al., 2017

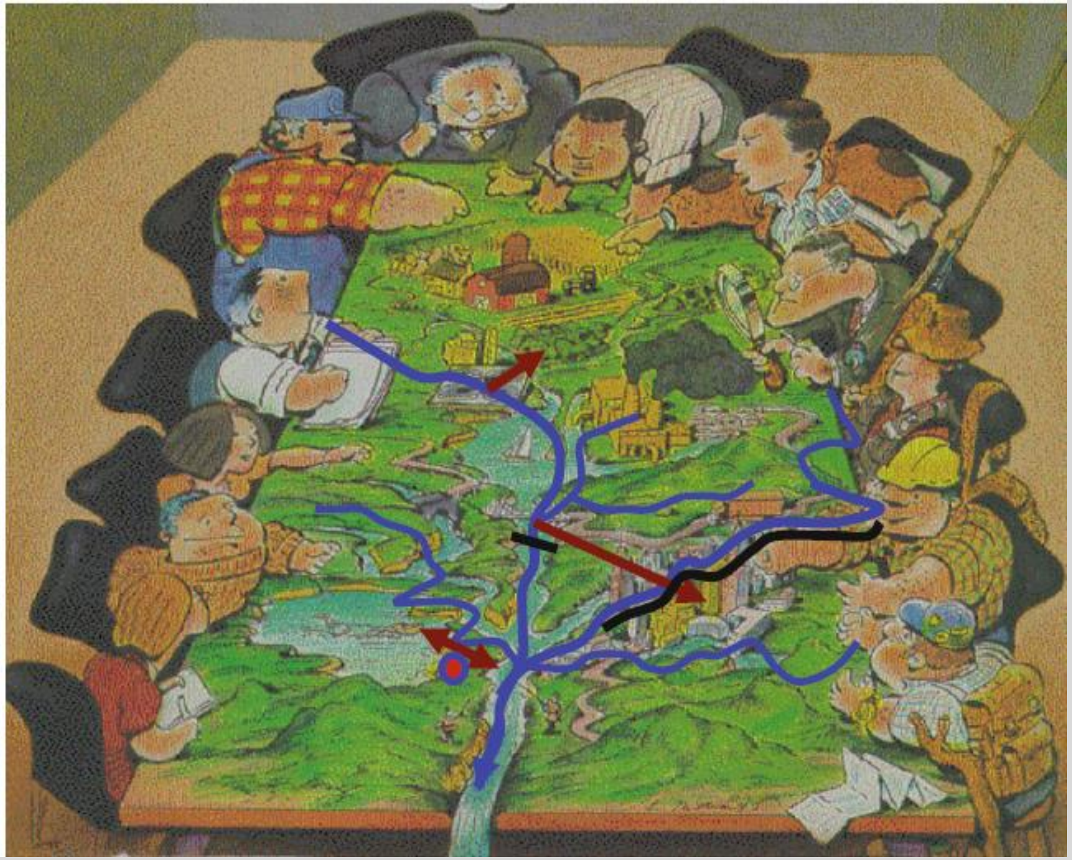
Planning, design and management is of importance to numerous stakeholders

identification and implementation of both structural and nonstructural measures designed to e.g.

- increase the reliability and decrease the cost of municipal, industrial, and agriculture water supplies,
- to decrease risks associated with droughts and floods,
- to improve water quality,
- to provide for commercial navigation,
- to provide for recreation,
- to enhance aquatic ecosystems,
- to produce hydropower,
- to sustain ecosystem services
- etc.

as appropriate for the particular river basin..

→ Planning, design and management relies on modelling.



Loucks et al., 2017

Different aims require different models

The purpose of modelling the transformation of precipitation into runoff is ***to simulate the response of a river basin to meteorological forcing***

- *to investigate the variability of hydrological processes and their impact on river flows*
- *to replace missing data, to extend historical data, to overcome the shortcomings due to limited measurements*
- *to predict river flows in ungauged basins*
- *to investigate scenarios*

→ **support engineering design, planning and management**

!!!Models are/must be characterised by different spatial and temporal representation of the transformation of precipitation into runoff depending on the aim of modelling!!!

1.2 Structure and Variables of a Hydrological Model

$$\mathbf{Z}_{\boldsymbol{sim}}(t) = G\,[\mathbf{X}(t),\ \mathbf{Y}(t),\ \mathbf{BC}(t),\ \boldsymbol{\theta}] + \boldsymbol{\varepsilon}_Z(t,\ \boldsymbol{\theta})$$

$\mathbf{Z}_{sim}(t)$: values of the various output variables of the hydrological model simulated at time t (system response)

$\mathbf{X}(t)$: input variables

$\mathbf{Y}(t)$: state variables

$\mathbf{BC}(t)$: boundary conditions at time t

$\boldsymbol{\theta}$**:** vector of model parameters

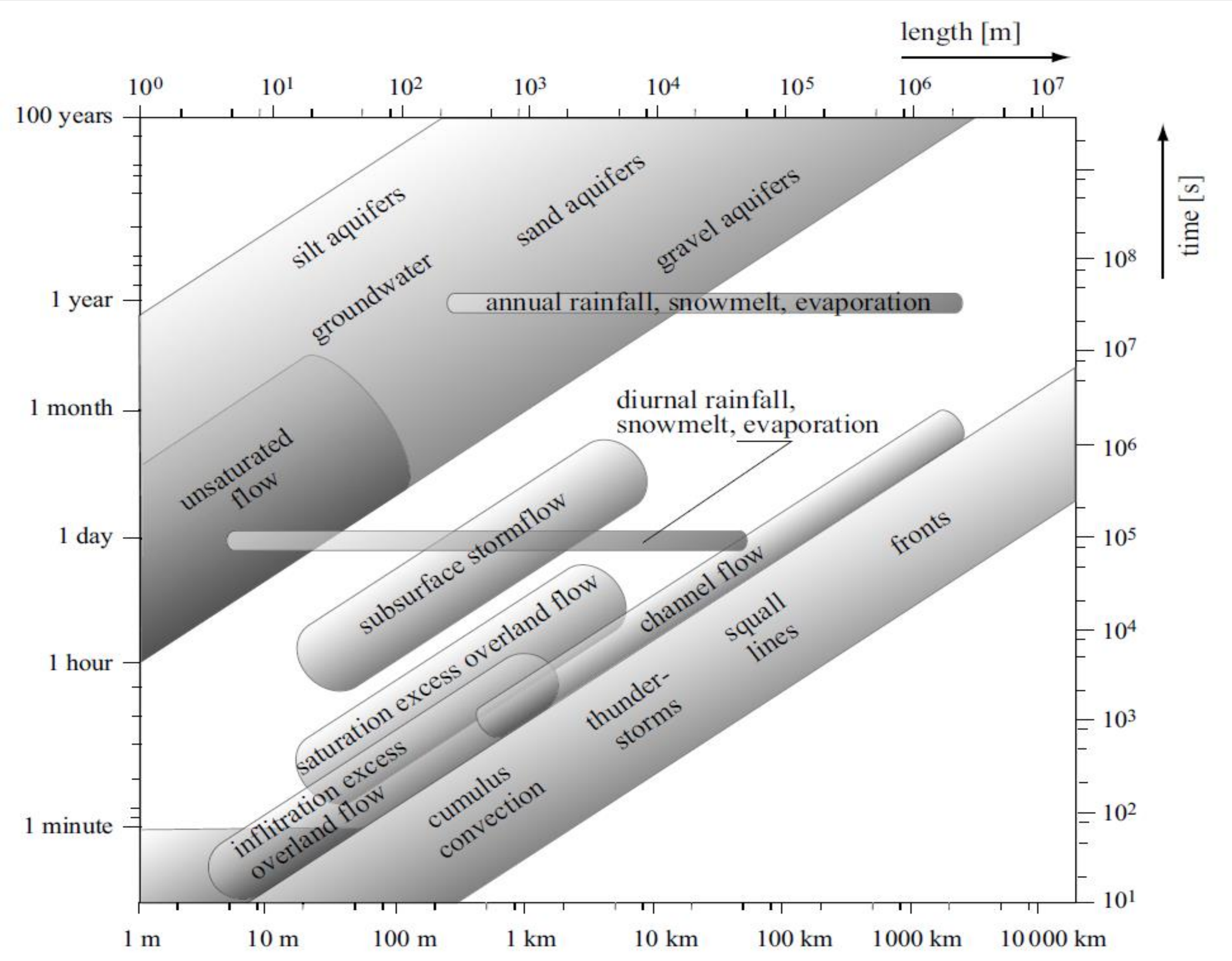
$\boldsymbol{\varepsilon}_Z(t,\ \boldsymbol{\theta})$: vector of model errors that are a function of time and parameter set $\boldsymbol{\theta}$

Factors affecting the hydrological system response

- Rainfall intensity & duration; snow cover & melt
- Catchment size
- Catchment slope
- Catchment shape
- Catchment storage
- Channel type and morphology
- Channel storage
- Land use/land cover
- Soil type & infiltration
- Imperviousness, infrastructure
- etc.

→ Relevant factors need to be identified and considered for successful hydrological modelling

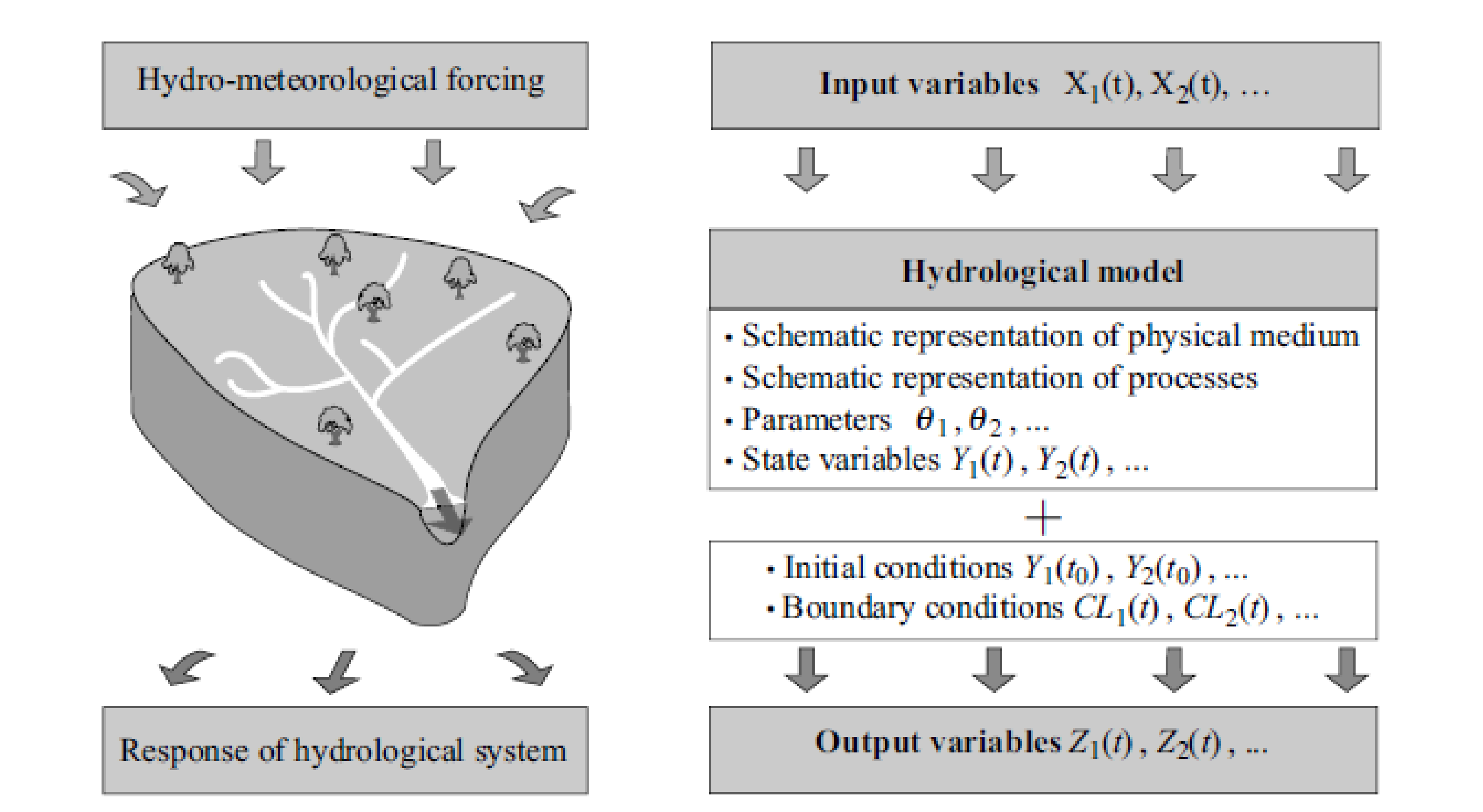
Temporal and spatial scales of various hydrological processes



Ref: Blöschl and Sivapalan (1995)

Typical structure and variables of a hydrological model

Model G $\mathbf{Z}_{sim}(t) = G\,[\mathbf{X}(t), \mathbf{Y}(t), \mathbf{BC}(t), \boldsymbol{\theta}] + \boldsymbol{\varepsilon}_Z(t, \boldsymbol{\theta})$



Hingray et al., 2015

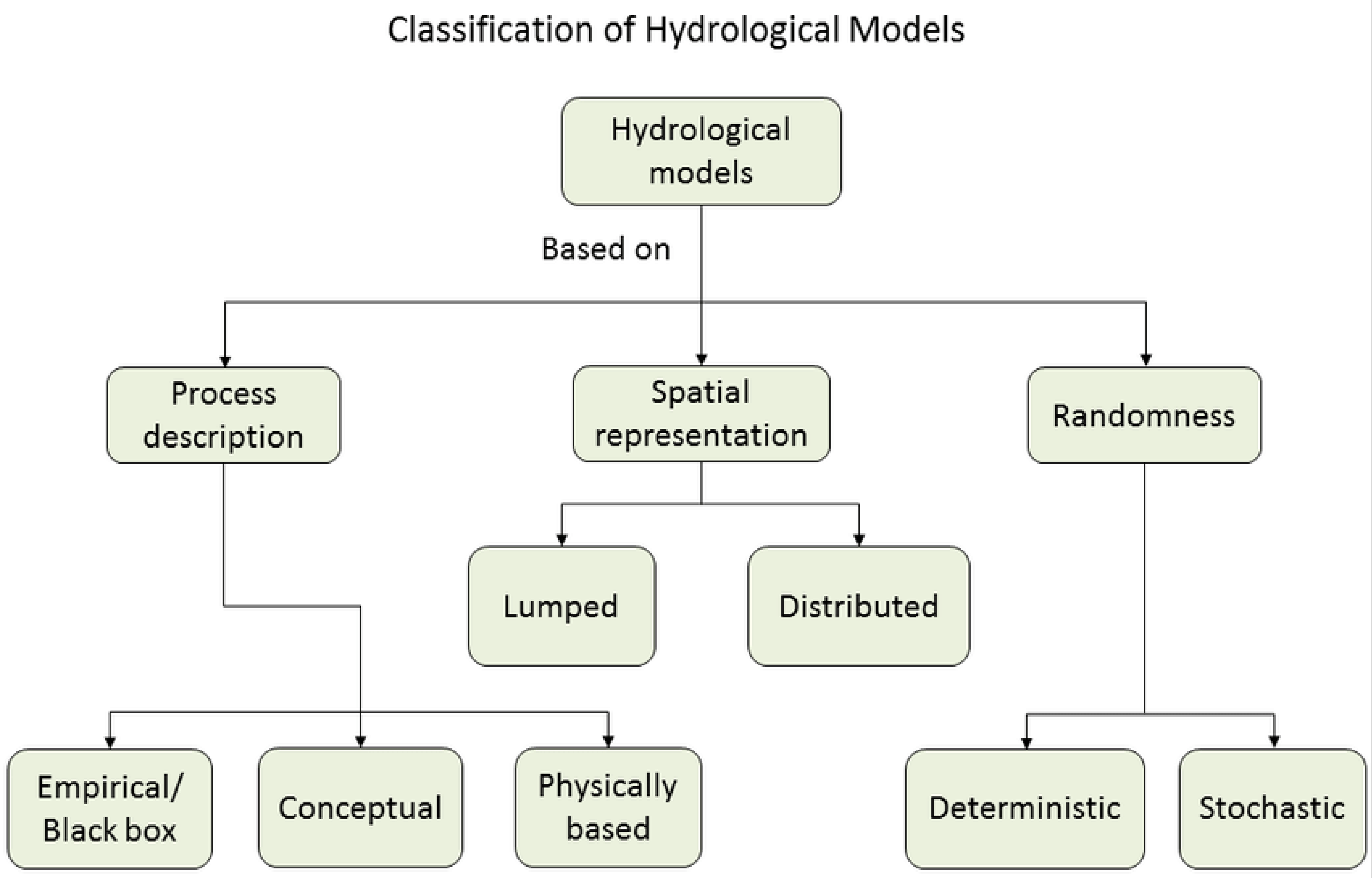
1.3 Main Types of Hydrological Models Model Classification

- **Physical models** – physical scale representation of real-world systems

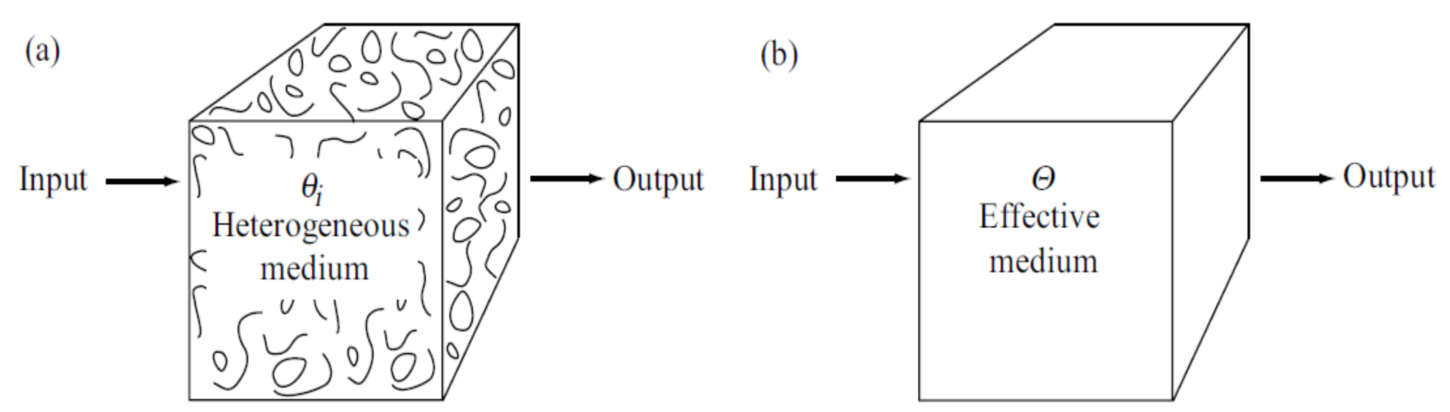
- **Numerical models** – equation or set of equations that represents the response of a real-world system (response of a hydrologic system component to a change in hydro-meteorological conditions)

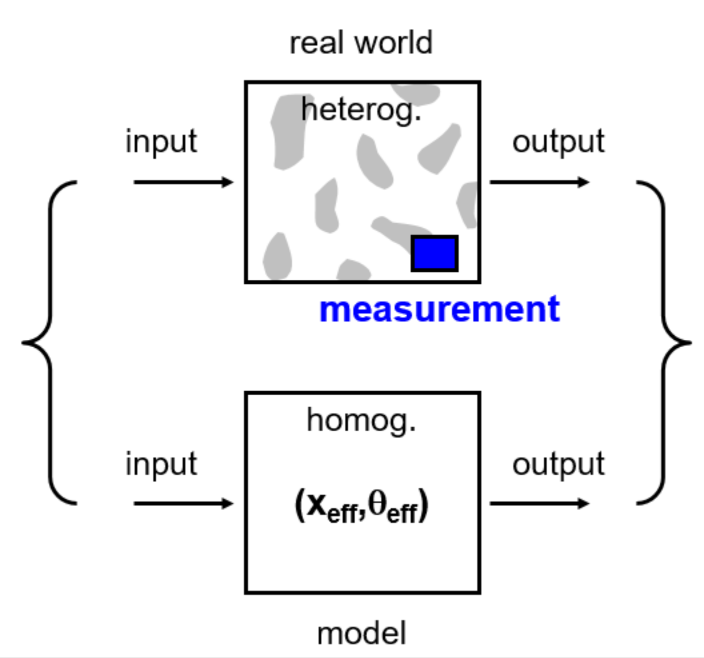
$$\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u} \cdot \nabla \boldsymbol{u} = -\frac{\nabla P}{\rho} + v\nabla^2 \boldsymbol{u}$$

1.3 Main Types of Hydrological Models



Representing the Processes





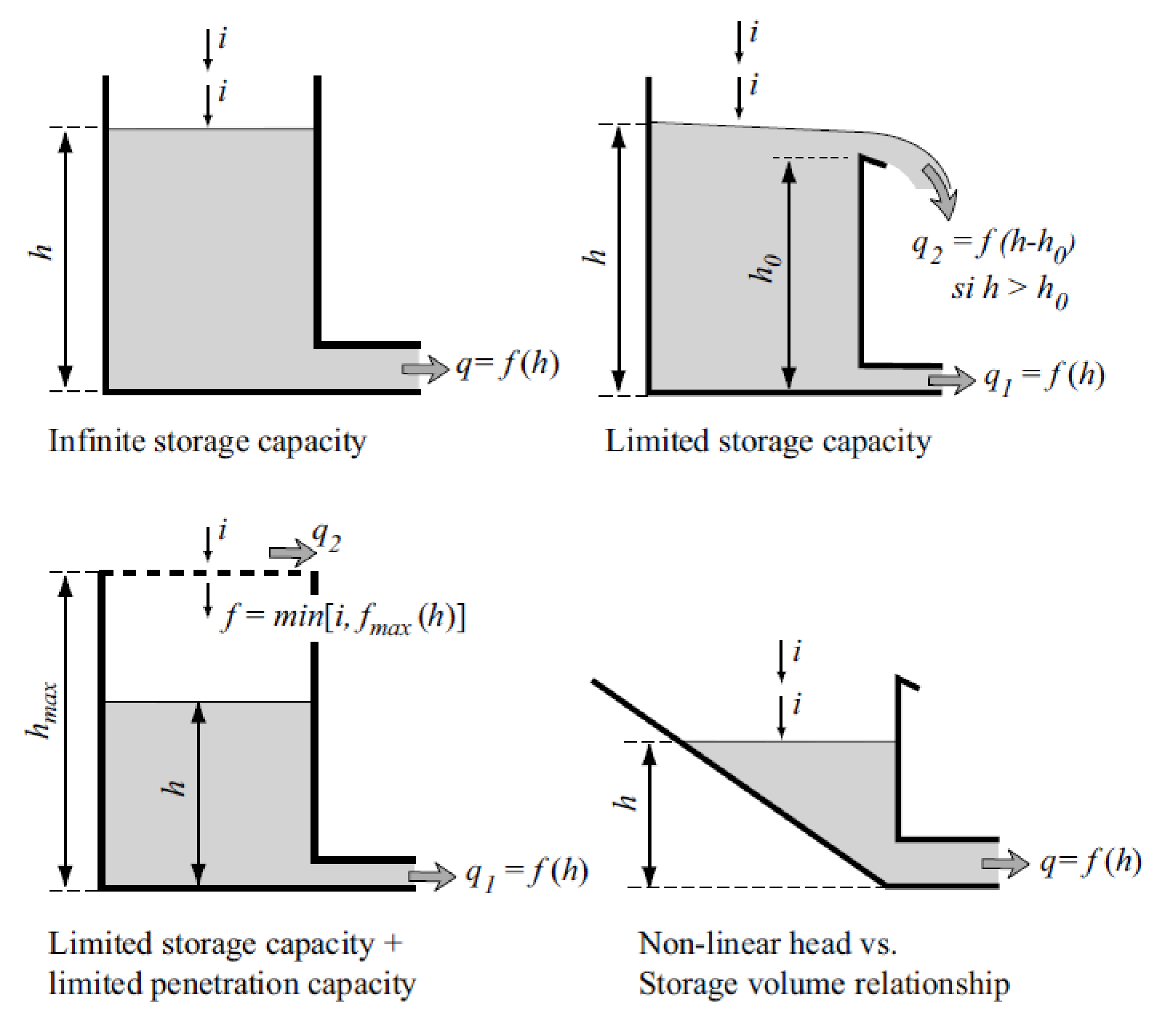
identical

Identical?

Ref:After Grayson and Blöschl, 2000, Cambridge Univ. Press

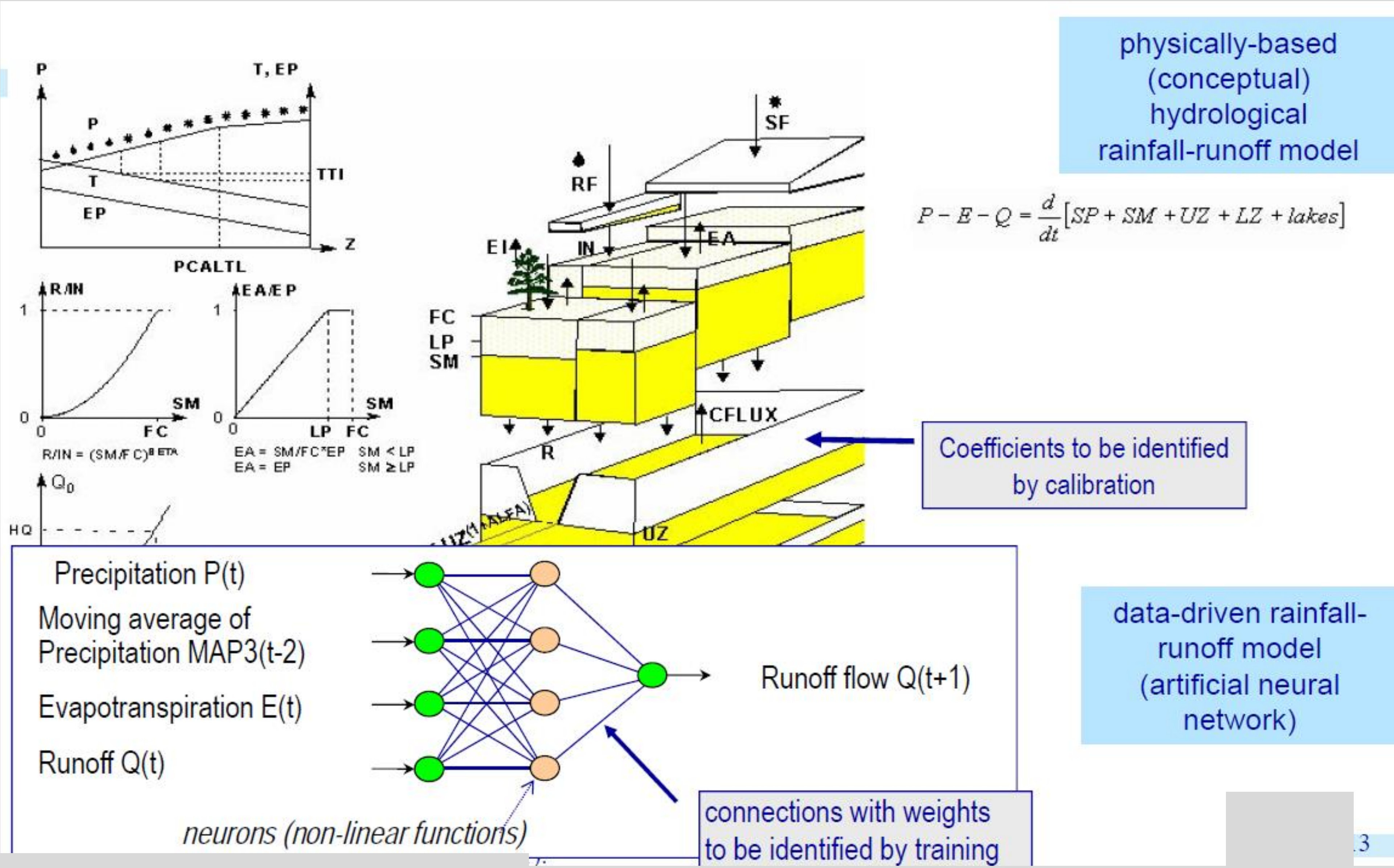
Classification: In-Confidence

Examples of conceptual reservoirs used in the structures of reservoir-type hydrological models



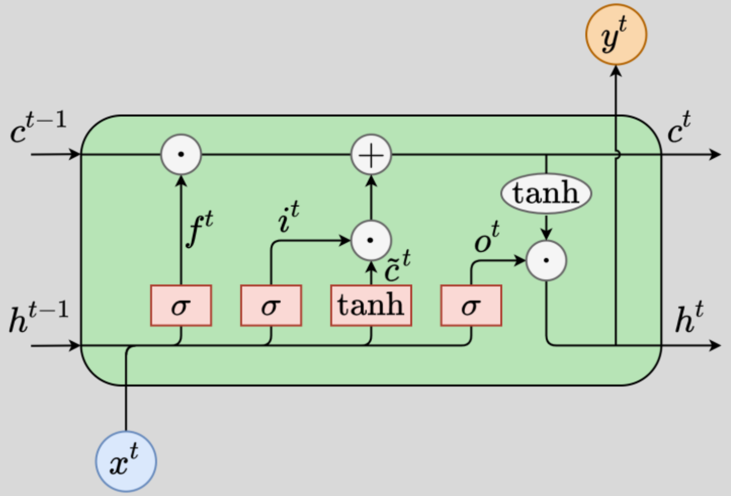
Hingray et al., 2015

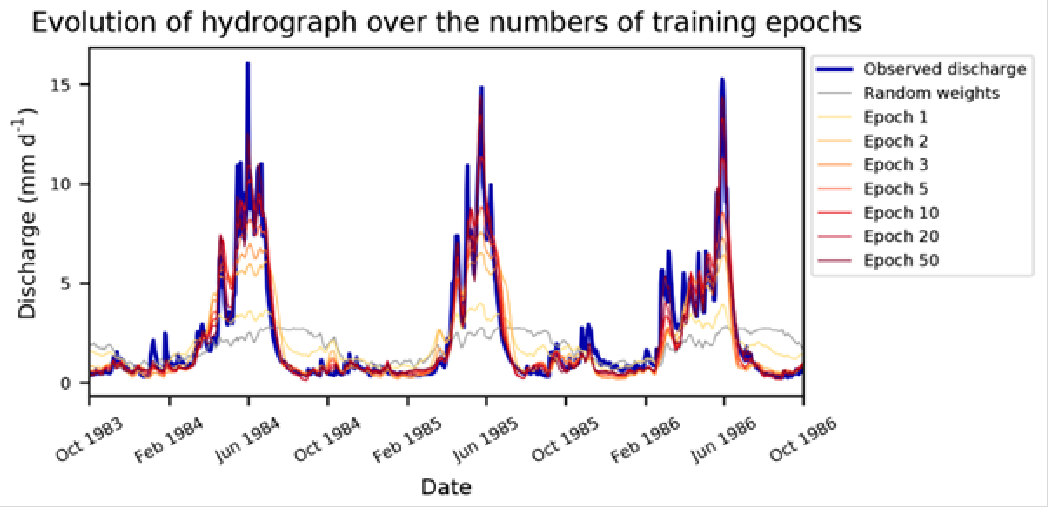
Data-driven vs Physically-based models

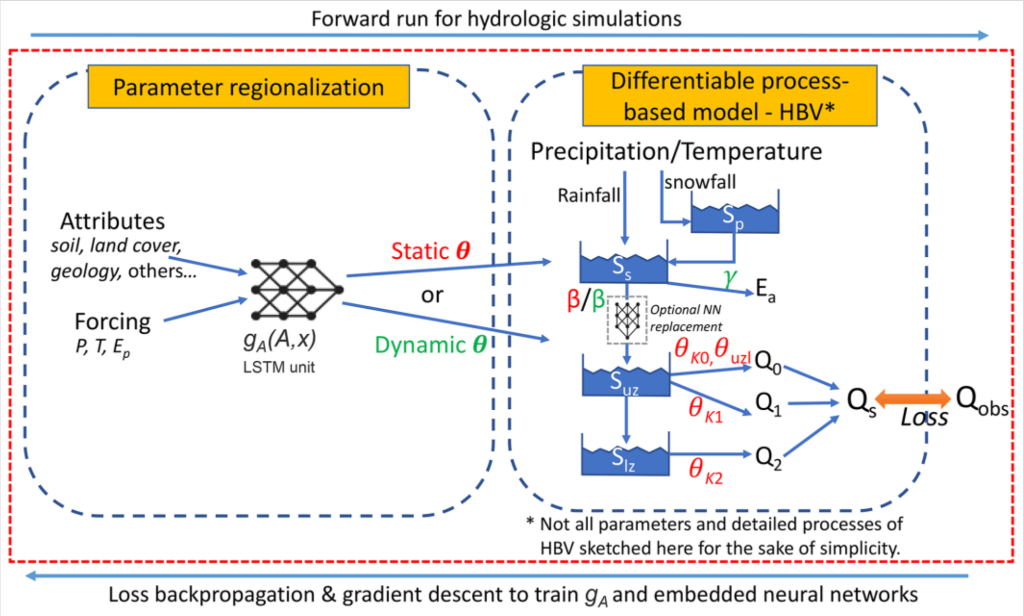


Machine Learning (see ENCN404)
Physics-informed Machine Learning

LSTM for rainfall-runoff modelling, presentation by Frederik Kratzert on YouTube: https://www.youtube.com/watch?v=skFAT8i2bZI







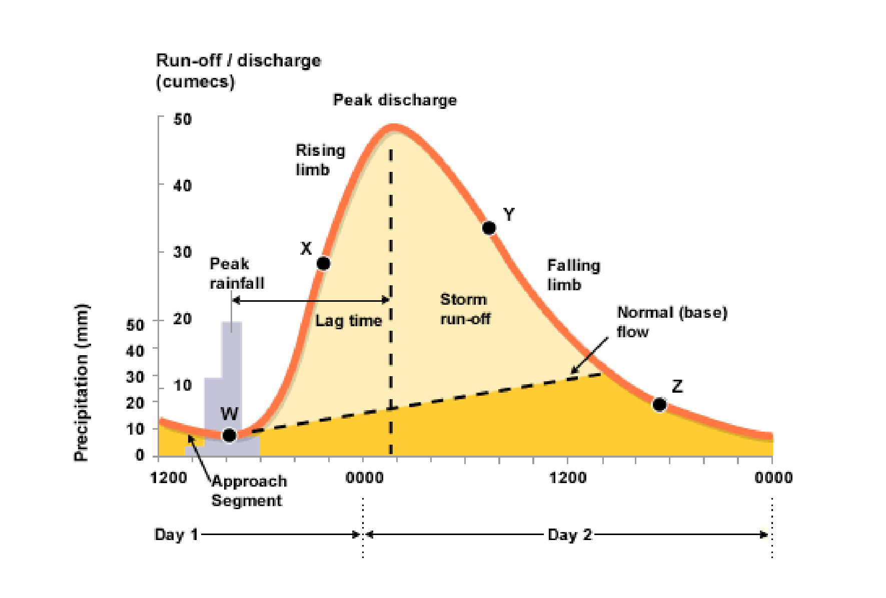
Kratzert et al. (2018)
Link to publication: https://hess.copernicus.org/articles/22/6005/2018/

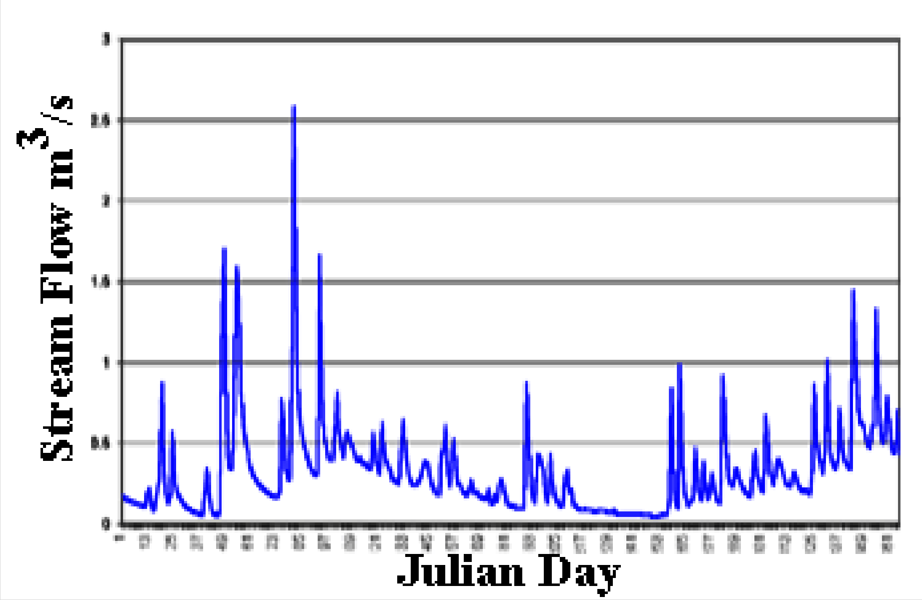
Feng et al. (2023)
Link to publication: https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2022WR032404

Classification of hydrological models

Scales (time factor)

- Event-based
- Continuous
- Hybrid





2 REPRESENTING THE PHYSICAL MEDIUM AND PROCESSES

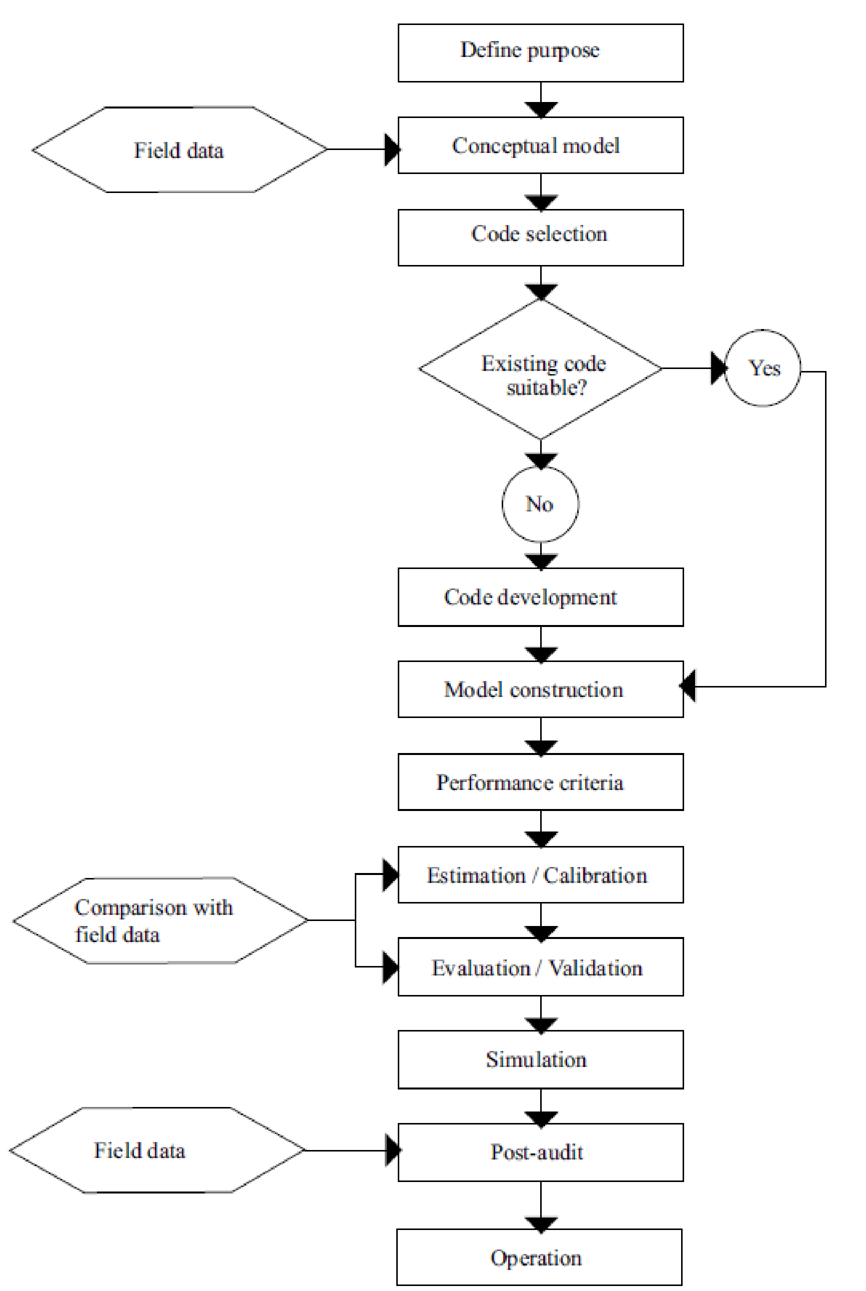
In this course the focus is on empirical, physically based and conceptual models.

Machine Learning models are covered in ENCN404 *Modern Modelling Practices for Civil Engineering* and in ENCI646 *Flood Analysis, Modelling and Management*.

2.1 Development of a Hydrological Model

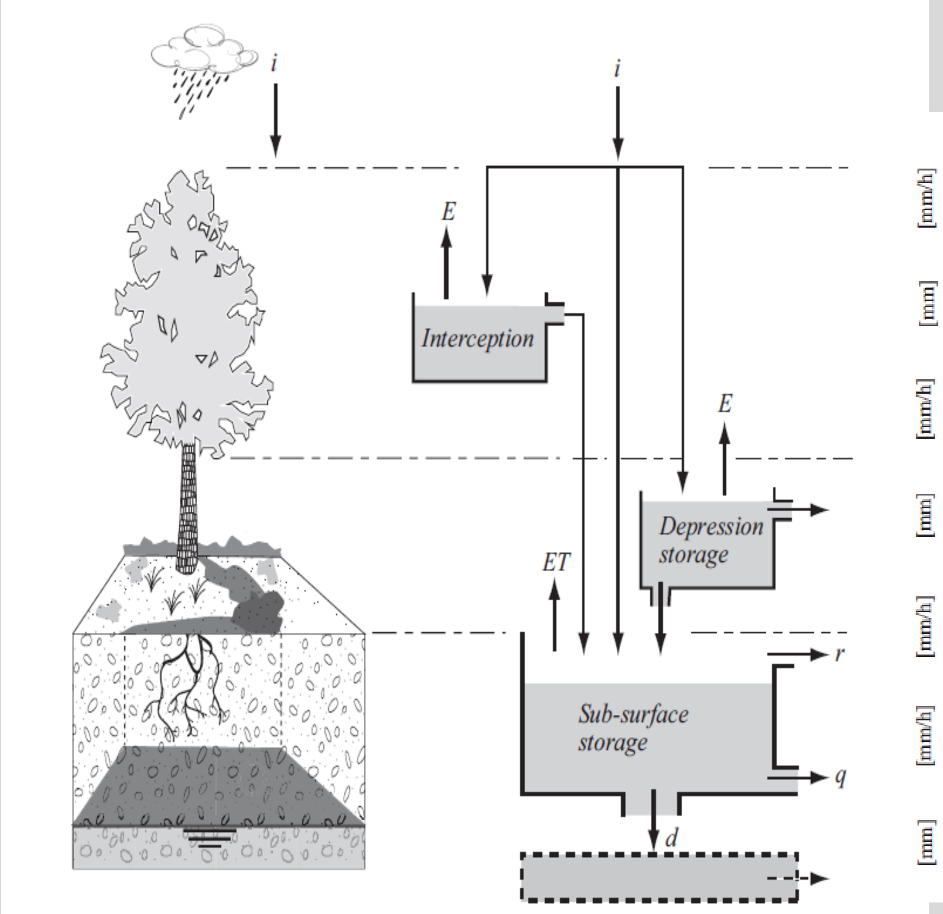
In brief:

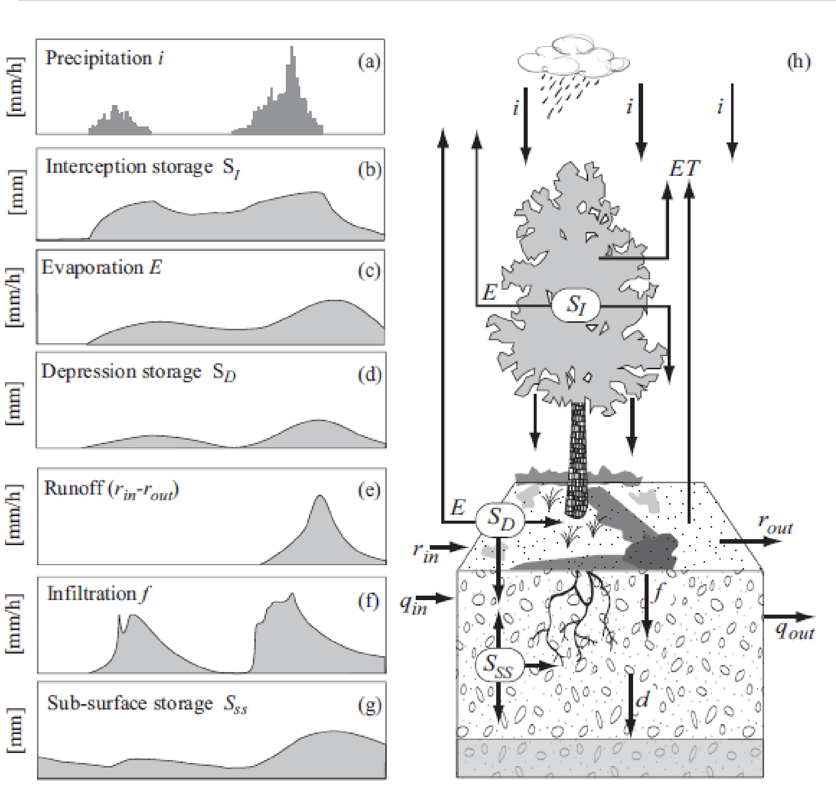
- Model construction
- Model calibration
- Model evaluation, in particular its validation



Hingray et al., 2015

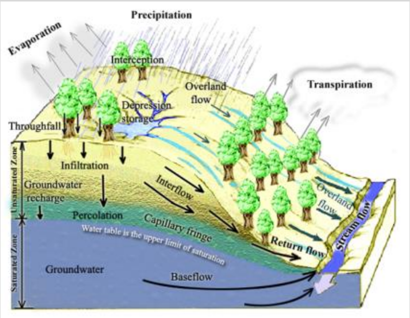
Example of a representation of the physical system as a hydrological model

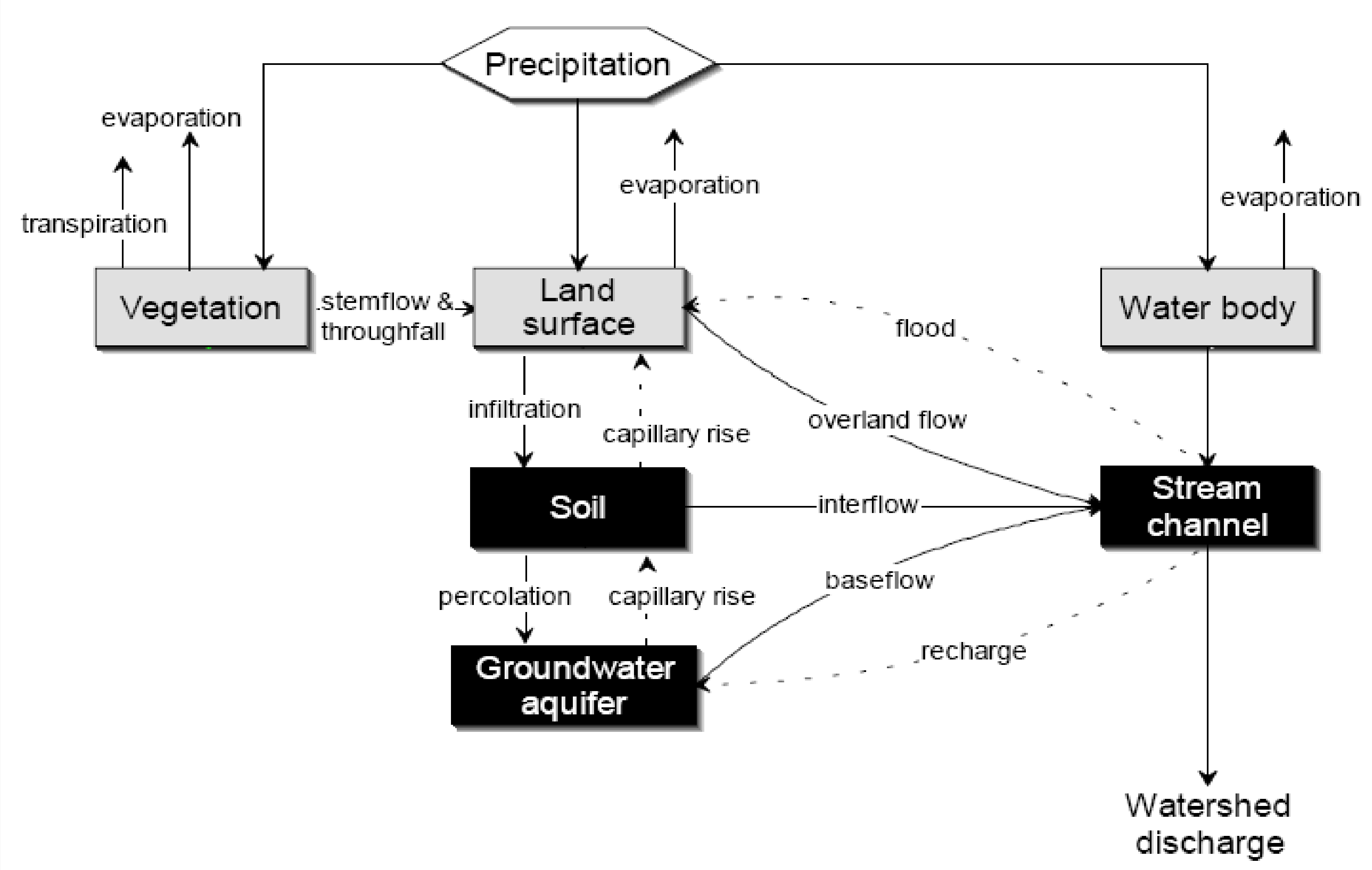




Hingray et al., 2015

2.2 Representing the Processes





https://www.hec.usace.army.mil/software/hec-hms/

Exercise 1

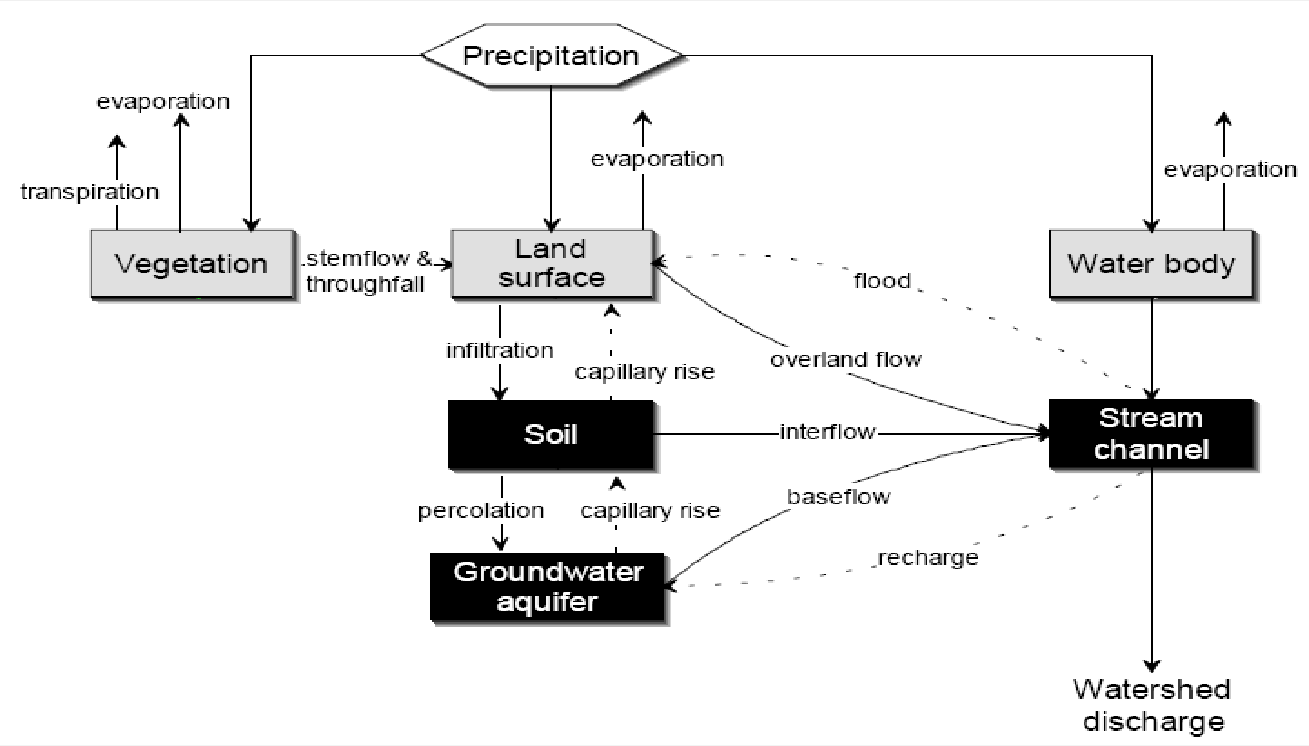
Construct a hydrological model concept by selecting an adequate process representation for the processes shown in the Figure.

Make sure to define the purpose of the hydrological model first.

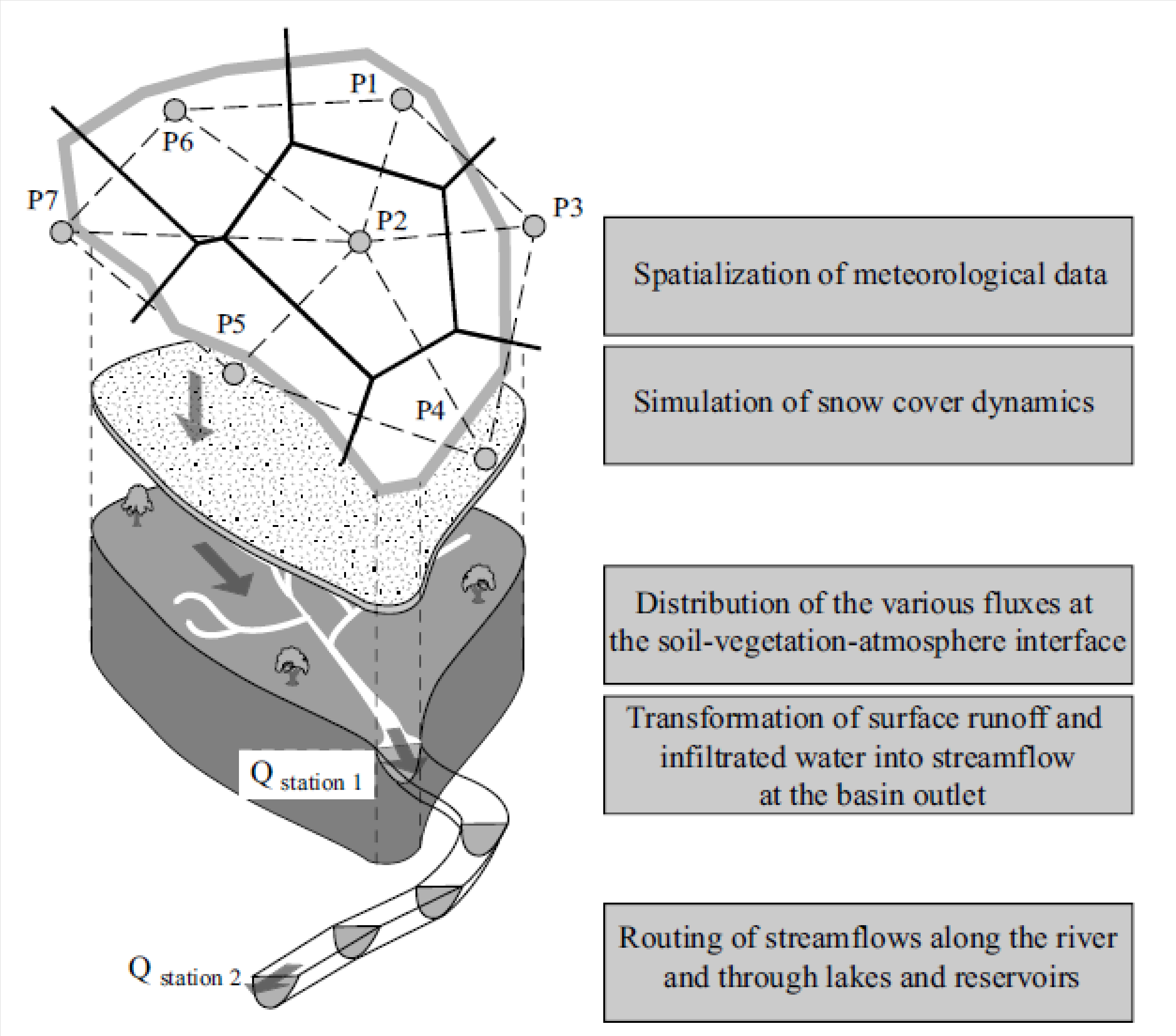
How would your model concept change if the purpose changes?

Discuss required model parameters and potential sources for these model parameters.

Which data are needed to apply the model that you have constructed?



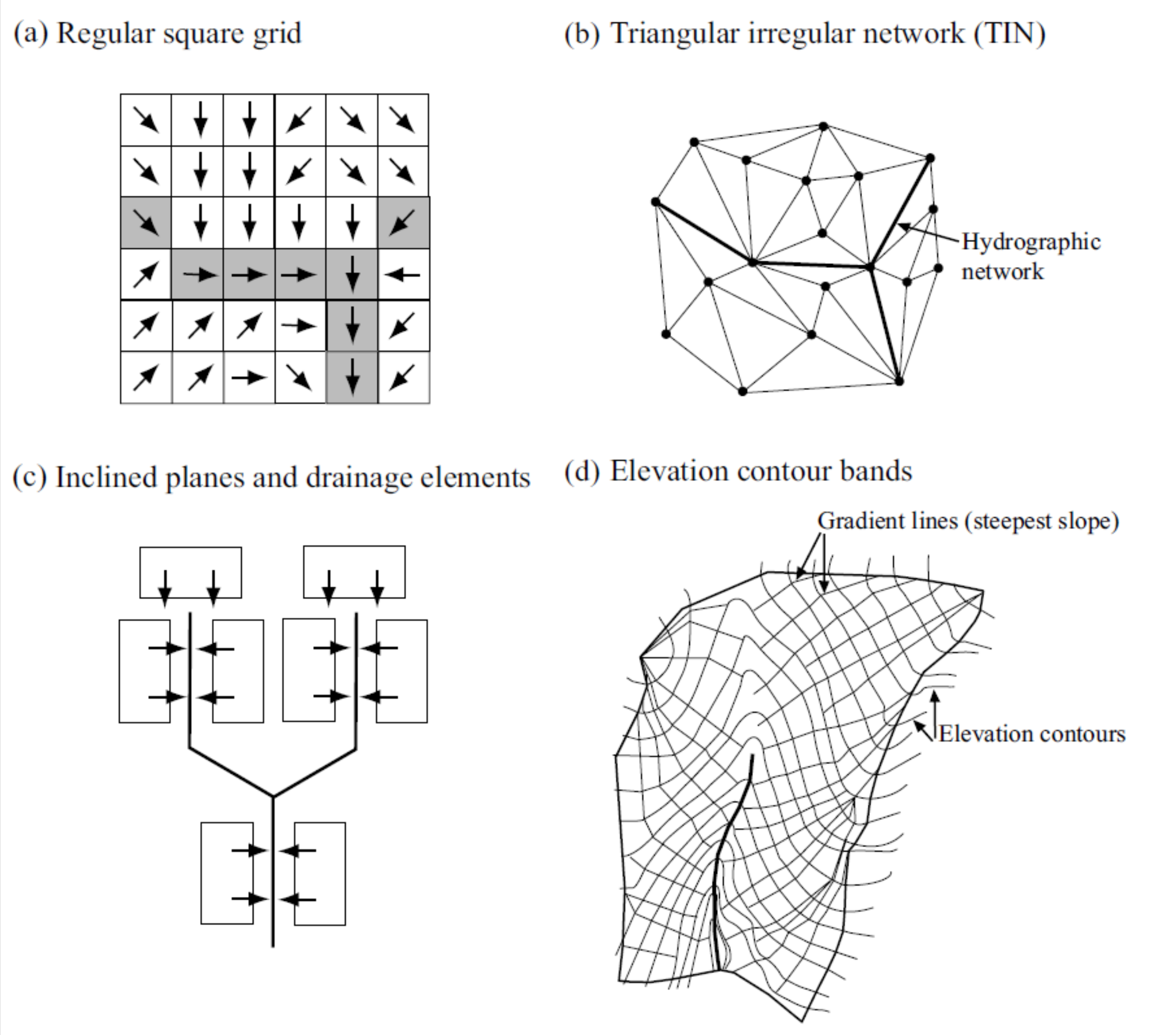
Typical steps of setting up and running a deterministic hydrological model



→ Computer lab

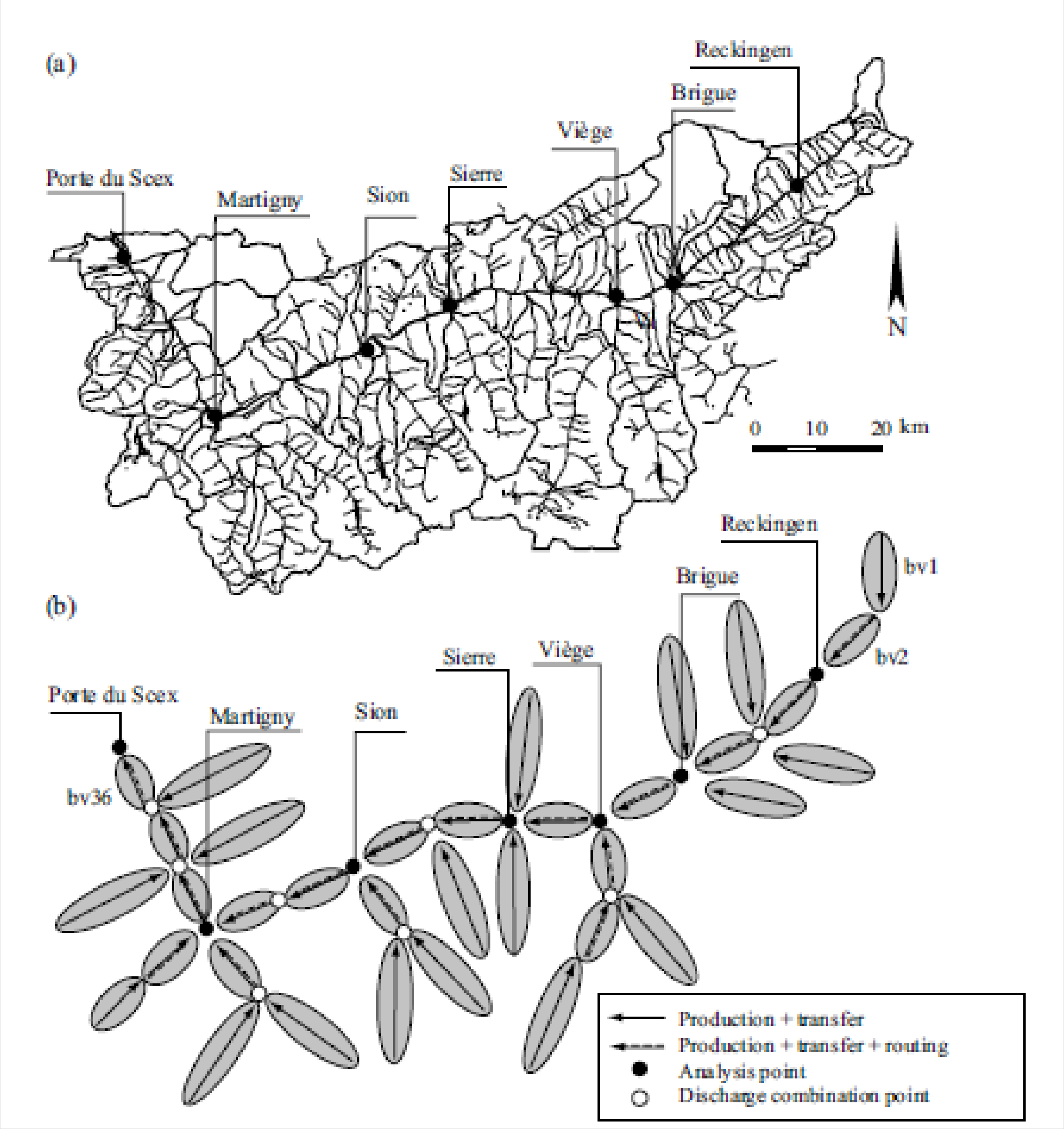
Hingray et al., 2015

Different terrain discretisation schemes used for hydrological models



(modified from Grayson *et al.*, 2000)

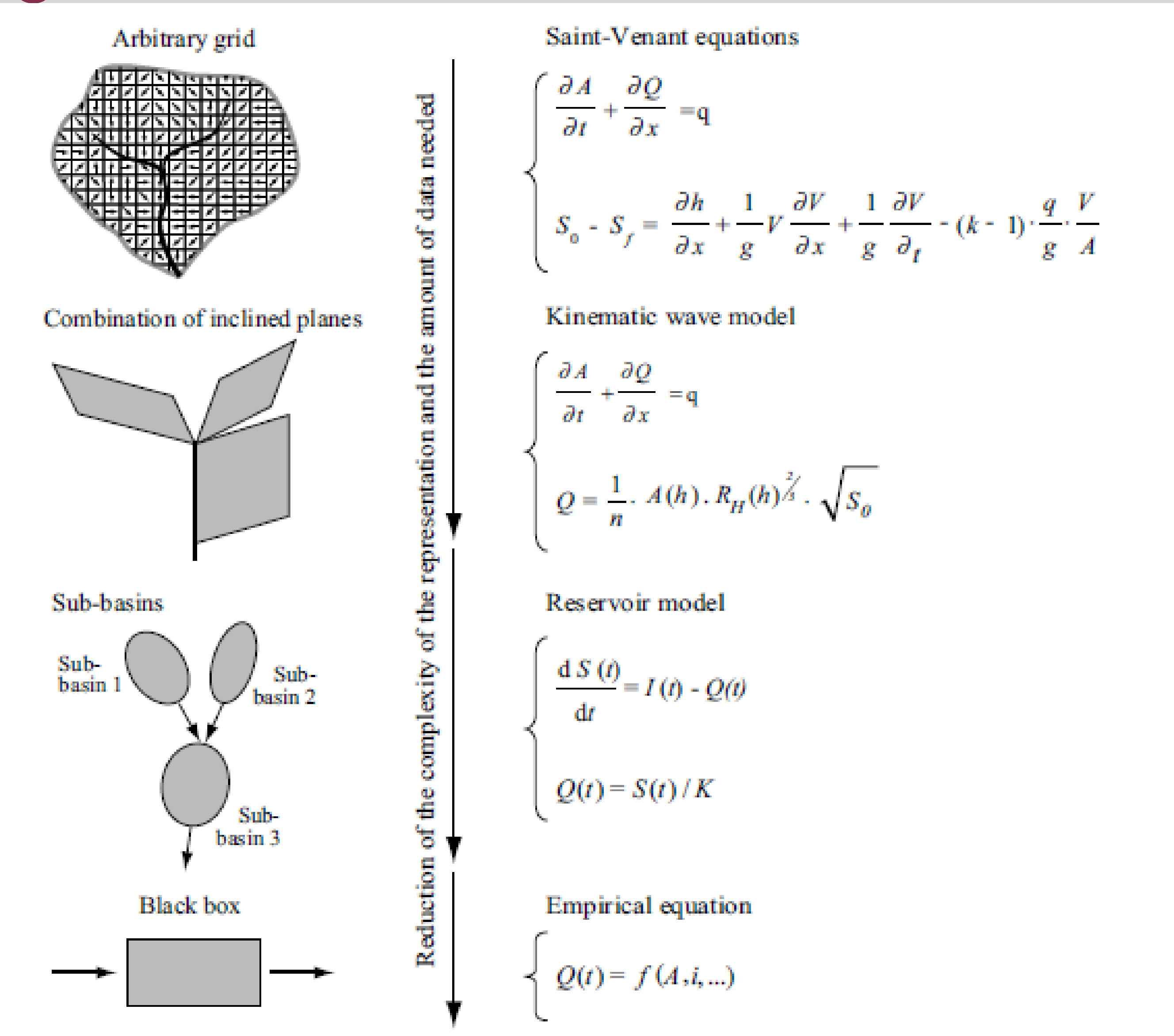
Example: Delineation of the Rhone drainage basin upstream of Lake Geneva into sub-basins



Hingray et al., 2015

2.4 Compatibility Between Spatial Representations and Processes

Different degrees of simplification that can be used when representing the geometry of the drainage basin and the corresponding surface runoff



Hingray et al., 2015

3 ESTIMATING MODEL PARAMETERS

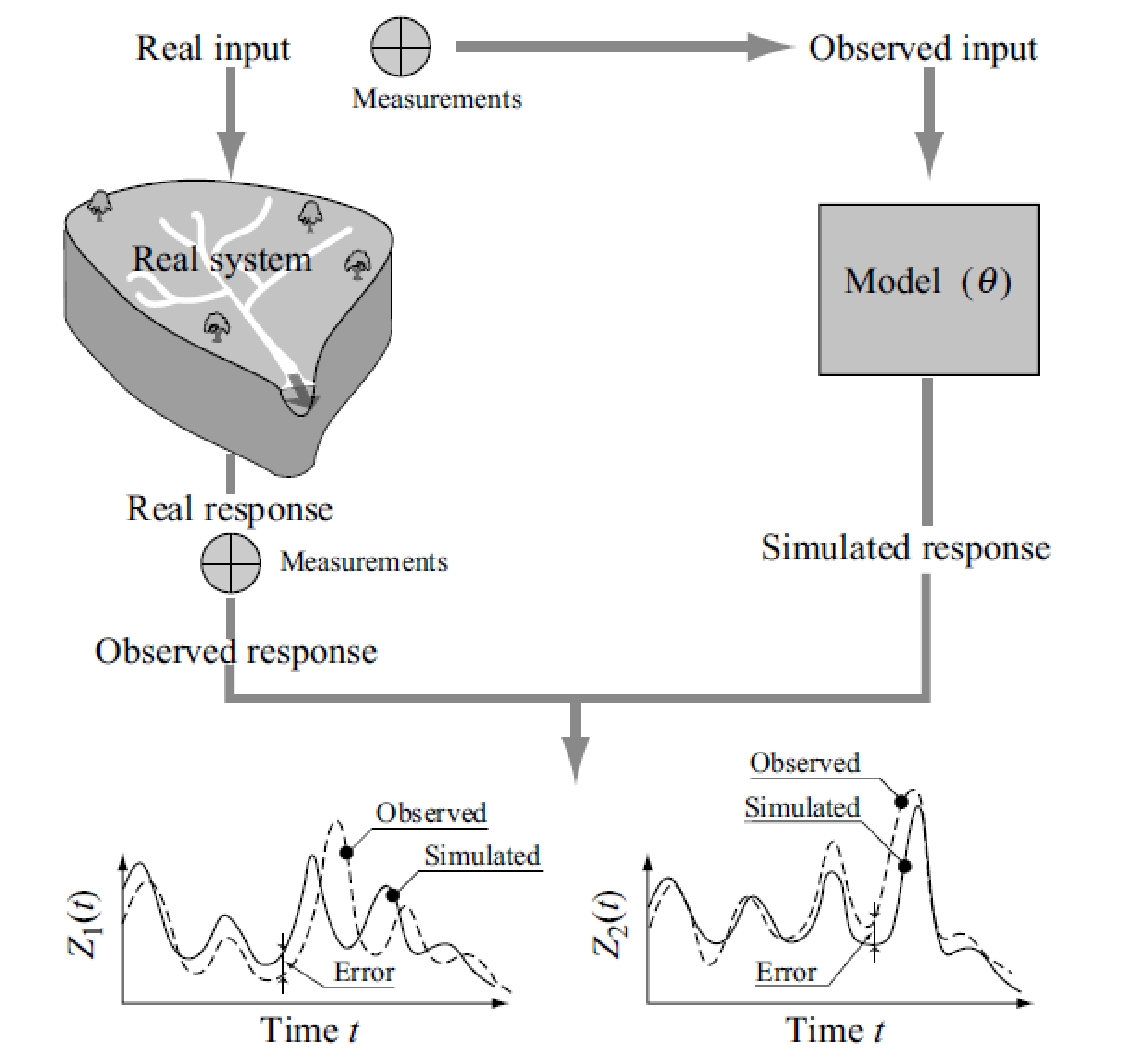
3.1 Nature of Parameters and Estimation Methods

- Approach 1:
  - Assigning values on the basis of **data** describing the drainage basin.
  - These data can be derived from measurements, field observations or thematic maps.
  - Often used to specify parameters that have a physical meaning. It can however be applied to estimate all types of parameters if appropriate regional models are available.
- Approach 2:
  - Estimation of the parameters by means of a **calibration** procedure.
  - This is carried out indirectly using hydrometeorological observations.
- Approach 3:
  - **Combination** of approaches 1 and 2

Procedure used to calibrate parameters θ of a hydrological model

$$\mathbf{Z}_{sim}(t, \boldsymbol{\theta}) = G\,[\mathbf{X}(t), \bar{\mathbf{Y}}(t), \mathbf{BC}(t), \boldsymbol{\theta}\,]$$

Calibration involves **minimising the error** between the **observed and simulated values** of various model output variables (e.g., $Z_1(t)$ and $Z_2(t)$)



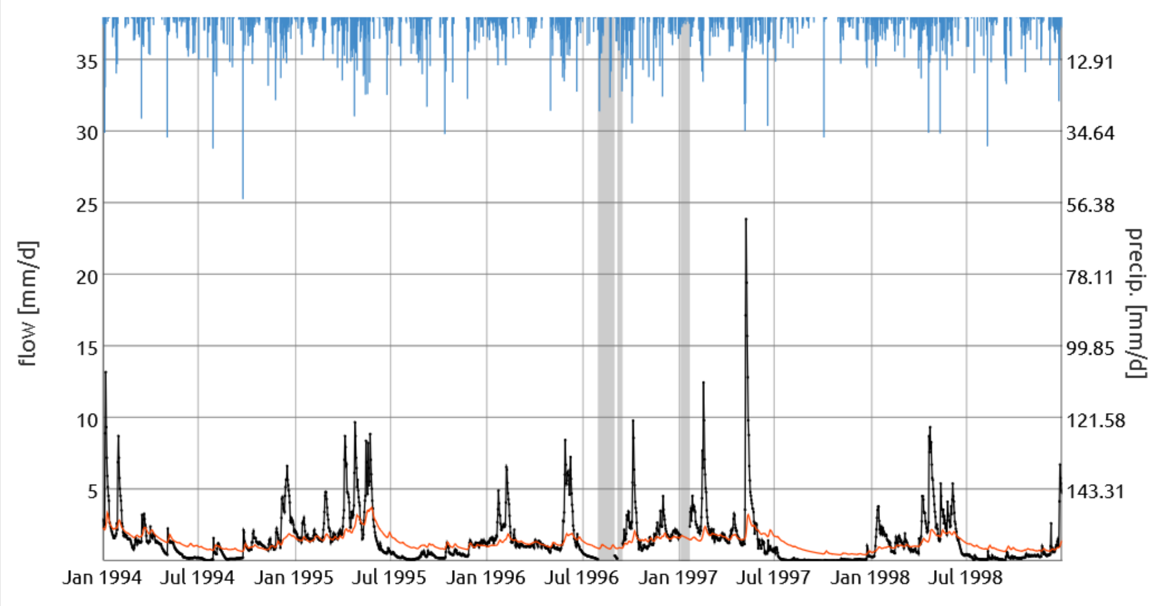
$$\varepsilon_Z(t,\boldsymbol{\theta}) = Z_{obs}(t) - Z_{sim}(t,\boldsymbol{\theta})$$

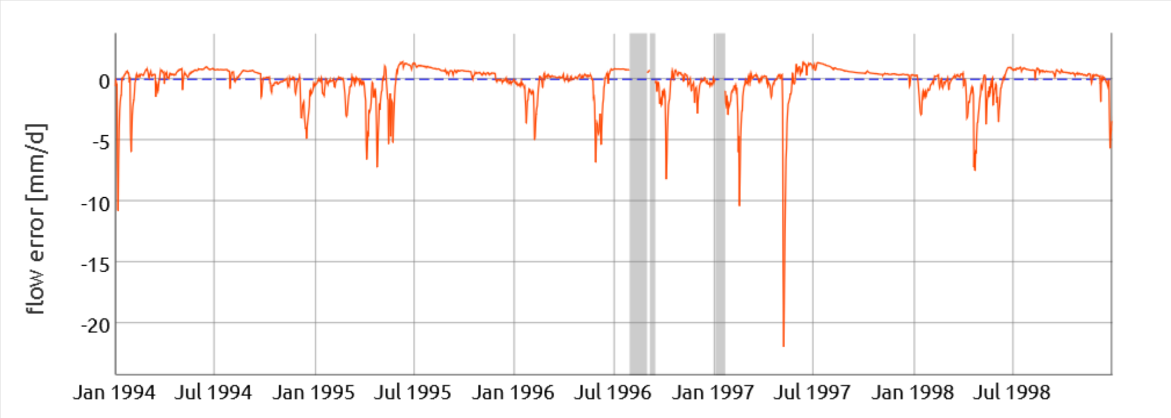
Hingray et al., 2015

3.2 Performance and Equifinality Criteria

- Model performance criteria
  - Using various **graphs** (e.g. time series plot, scatter plot, flow duration curve, etc.)
  - Quantitative assessment using an **objective function**
- Equifinality criteria
  - A number of parameter sets can generally be found to give **equivalent model performance** for a selected performance criterion.
  - Questions the concept of an optimal parameter set and introduces what is commonly referred to as the ***equifinality problem***.
  - This problem **increases** with the **number of parameters**.

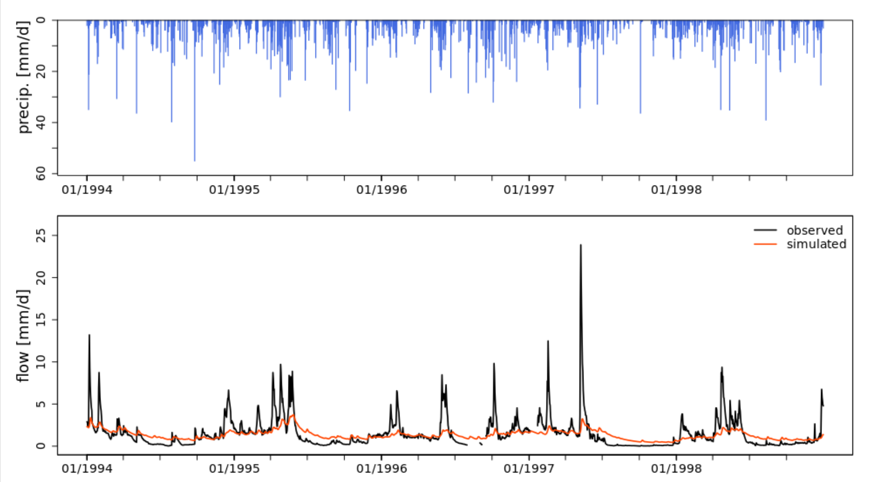
Performance Criteria - graphical

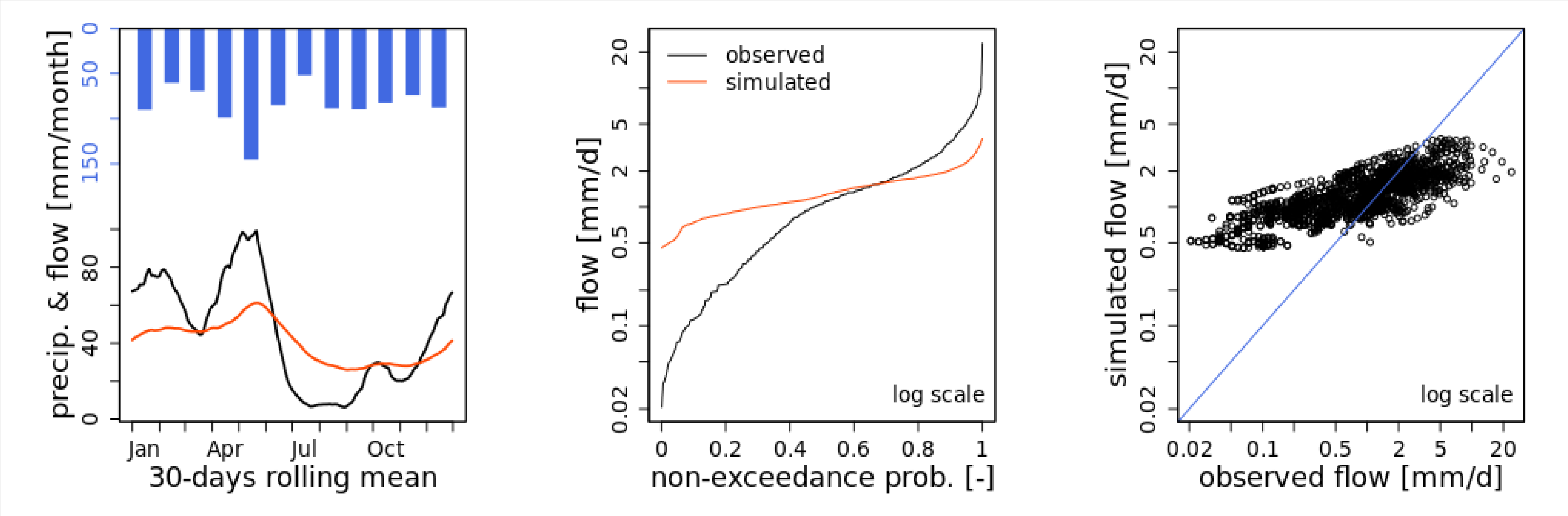




https://sunshine.inrae.fr/app/airGRteaching

Performance Criteria graphical





https://sunshine.inrae.fr/app/airGRteaching

Performance Criteria - quantitative

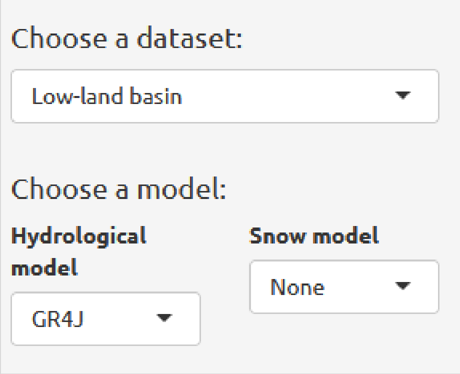
- Abbreviation, definition, range and optimum value of selected performance criteria (objective functions)

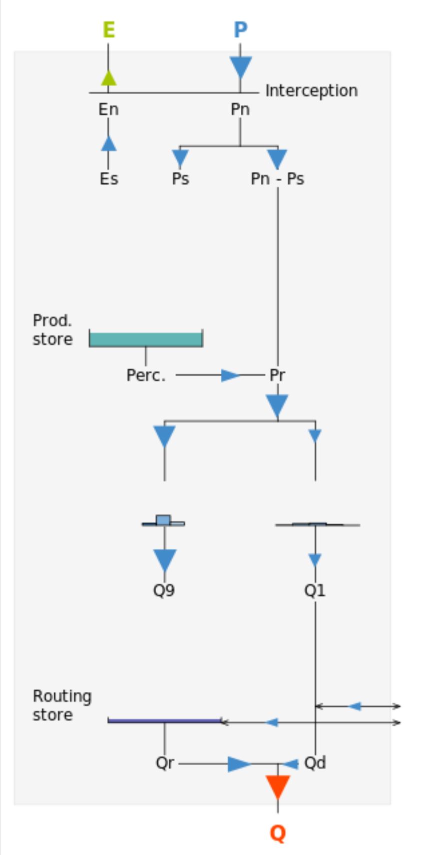
NSE	$1-\frac{\sum_{i=1}^{n}(O_i-P_i)^2}{\sum_{i=1}^{n}(O_i-\bar{O})^2}$	-∞ to 1.0	1.0
PBIAS	$\frac{\sum_{i=1}^{n}O_i-P_i}{\sum_{i=1}^{n}O_i}\times 100$	-∞ to ∞	0.0

Morias et al., 2015

Exercise 3

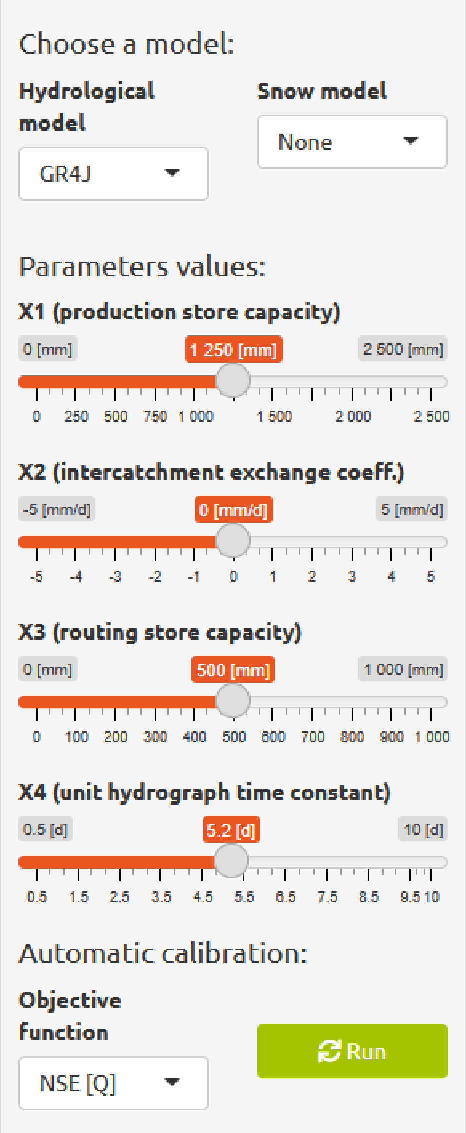
- Use airGRteaching (https://sunshine.inrae.fr/app/airGRteaching) to explore manual and automatic calibration of a hydrological model. Also explore using different types of plots to assess the model calibration exercise.
- Start with GR4J without snow model for a low-land basin.
- Explore other options:
  - Mountainous basin, without and with snow model
  - Different hydrological model (more parameters)
  - Other calibration objective than NSE
  - Plot previous simulation for comparison

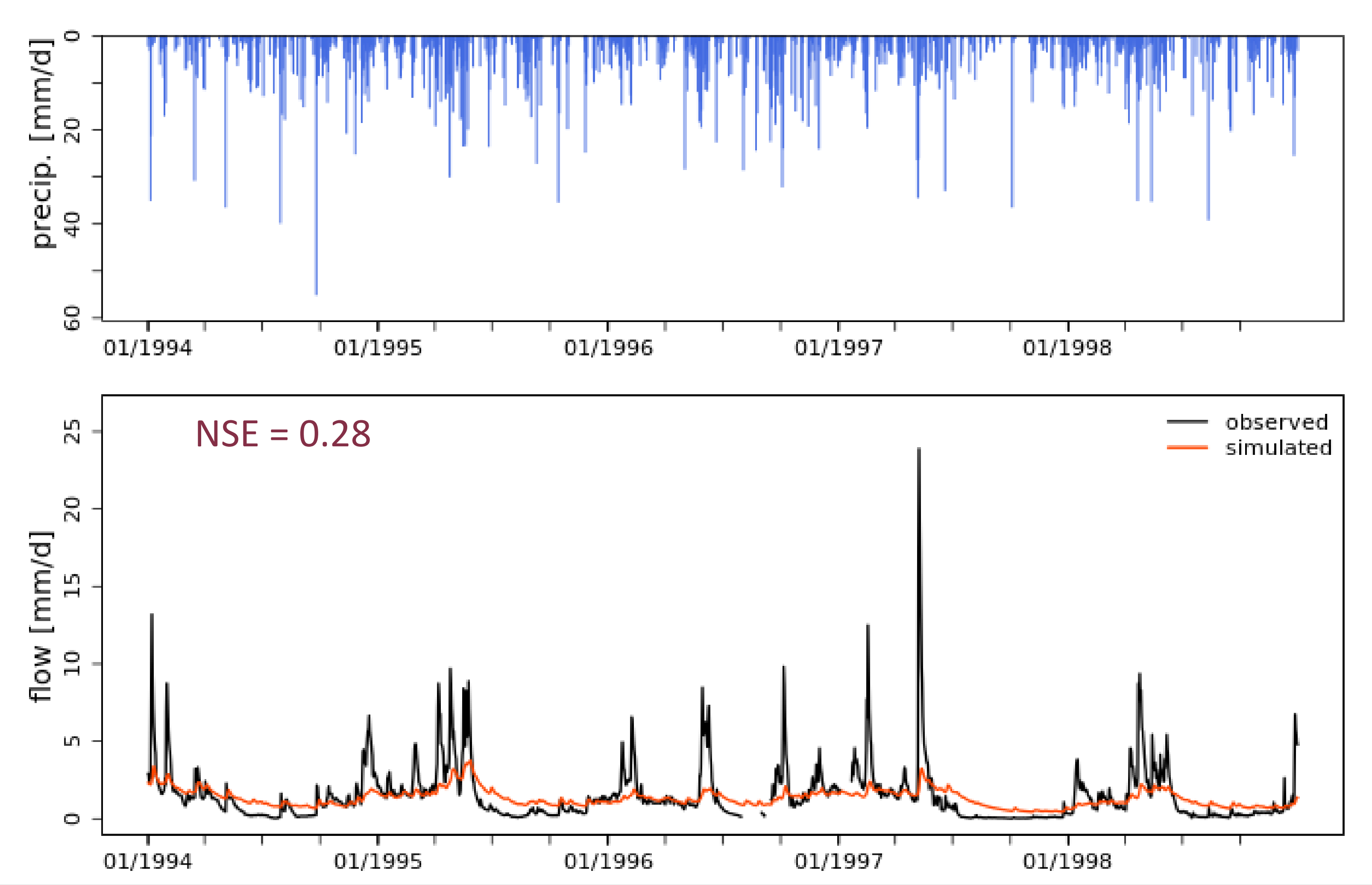




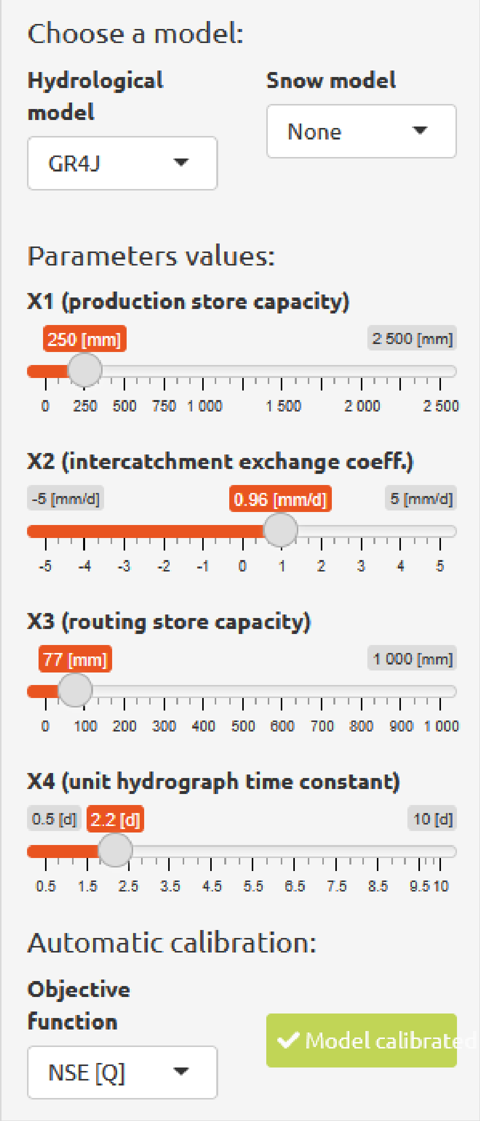
Main airGRteaching page:
https://hydrogr.github.io/airGRteaching/index.html

Exercise 3

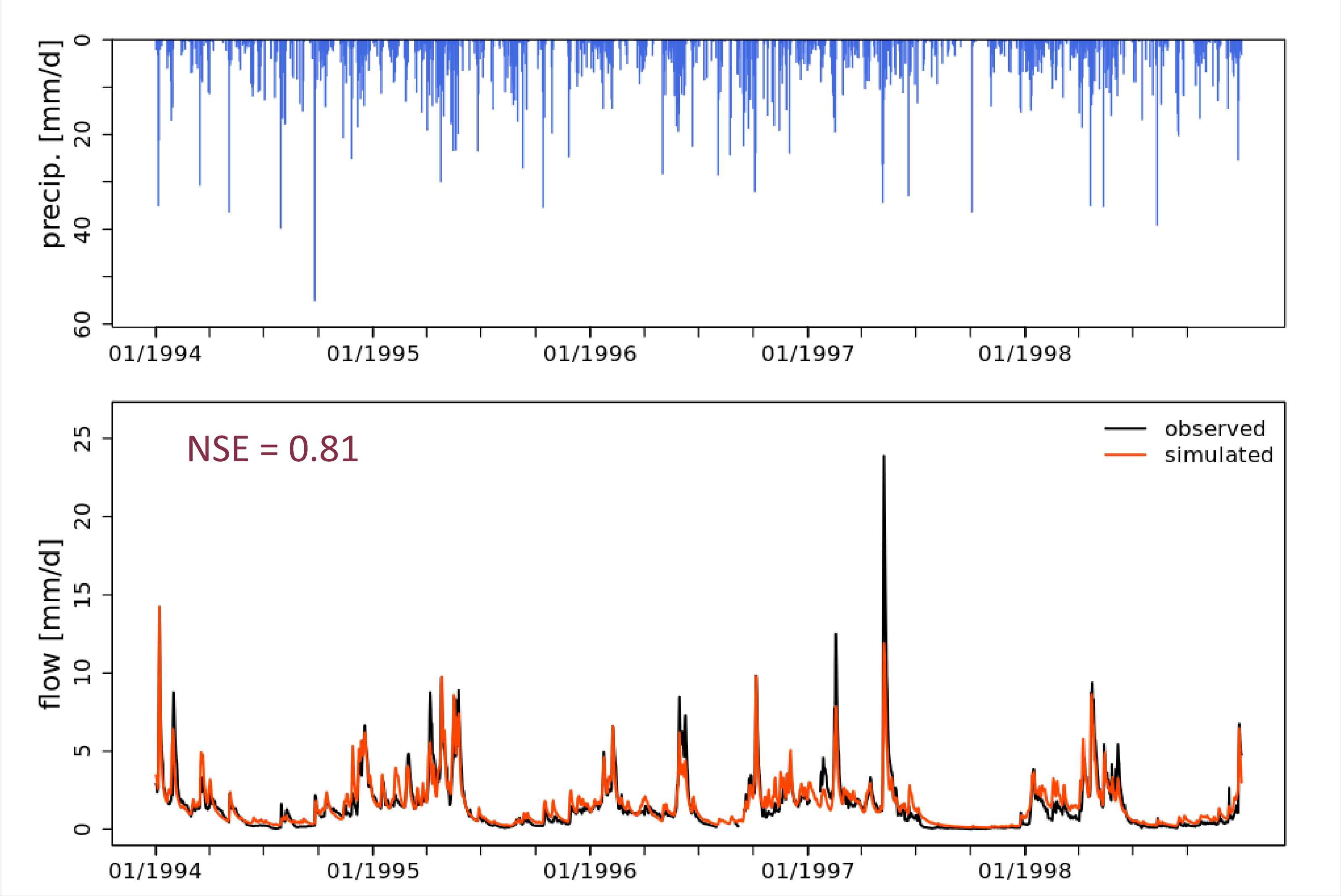

Choose a model:
Hydrological model
GR4J
Snow model
None
Parameters values:
X1 (production store capacity)
0 [mm]
1 250 [mm]
2 500 [mm]
0 250 500 750 1 000 1 500 2 000 2 500
X2 (intercatchment exchange coeff.)
-5 [mm/d]
0 [mm/d]
5 [mm/d]
-5 -4 -3 -2 -1 0 1 2 3 4 5
X3 (routing store capacity)
0 [mm]
500 [mm]
1 000 [mm]
0 100 200 300 400 500 600 700 800 900 1 000
X4 (unit hydrograph time constant)
0.5 [d]
5.2 [d]
10 [d]
0.5 1.5 2.5 3.5 4.5 5.5 6.5 7.5 8.5 9.5 10
Automatic calibration:
Objective function
NSE [Q]
Run


precip. [mm/d]
0
20
40
60
01/1994
01/1995
01/1996
01/1997
01/1998
NSE = 0.28
observed
simulated
flow [mm/d]
0
5
10
15
20
25
01/1994
01/1995
01/1996
01/1997
01/1998

Exercise 3

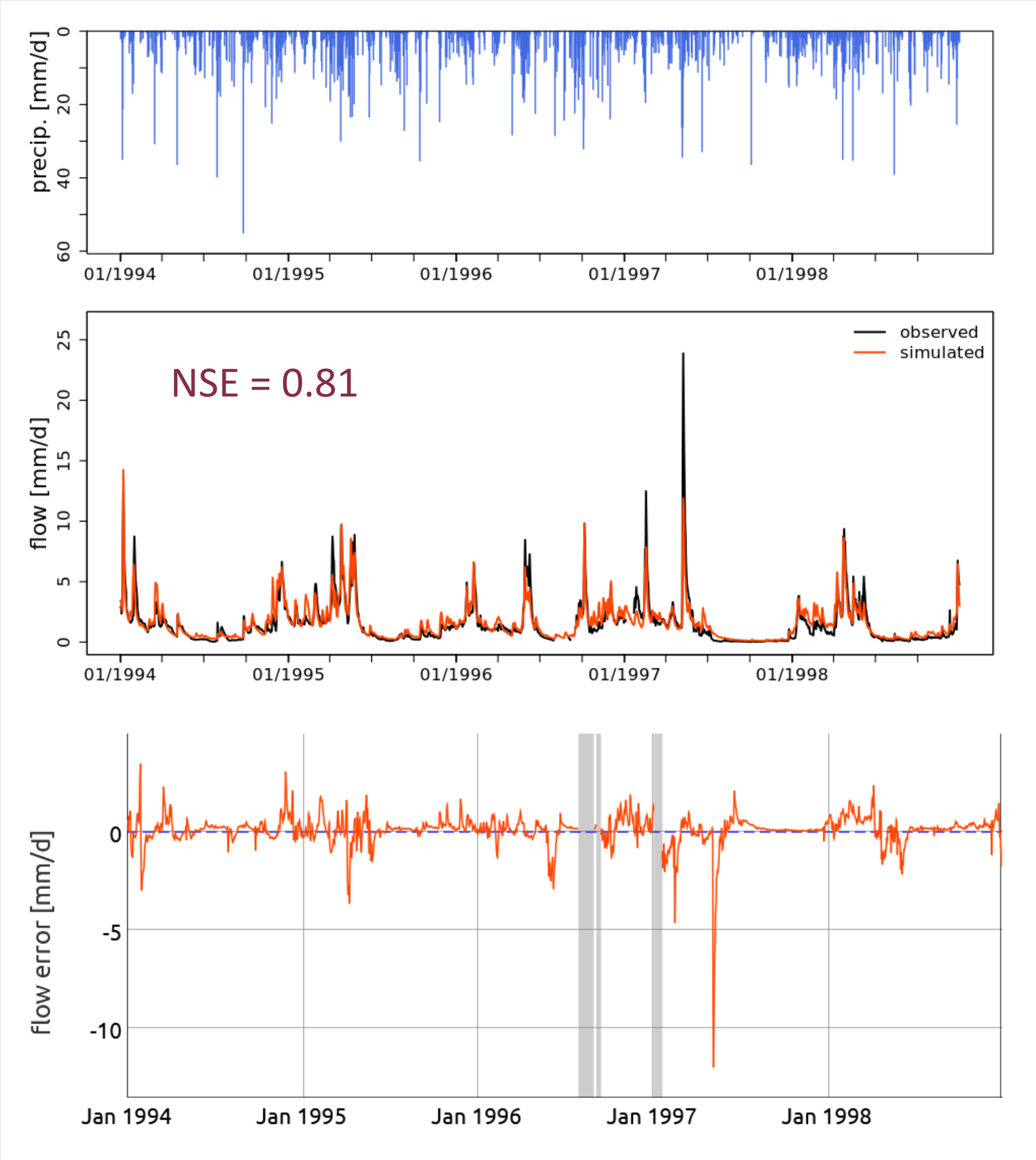


Choose a model:
Hydrological model
GR4J
Snow model
None
Parameters values:
X1 (production store capacity)
250 [mm]
2 500 [mm]
0 250 500 750 1 000 1 500 2 000 2 500
X2 (intercatchment exchange coeff.)
-5 [mm/d]
0.96 [mm/d]
5 [mm/d]
-5 -4 -3 -2 -1 0 1 2 3 4 5
X3 (routing store capacity)
77 [mm]
1 000 [mm]
0 100 200 300 400 500 600 700 800 900 1 000
X4 (unit hydrograph time constant)
0.5 [d]
2.2 [d]
10 [d]
0.5 1.5 2.5 3.5 4.5 5.5 6.5 7.5 8.5 9.5 10
Automatic calibration:
Objective function
NSE [Q]
Model calibrate



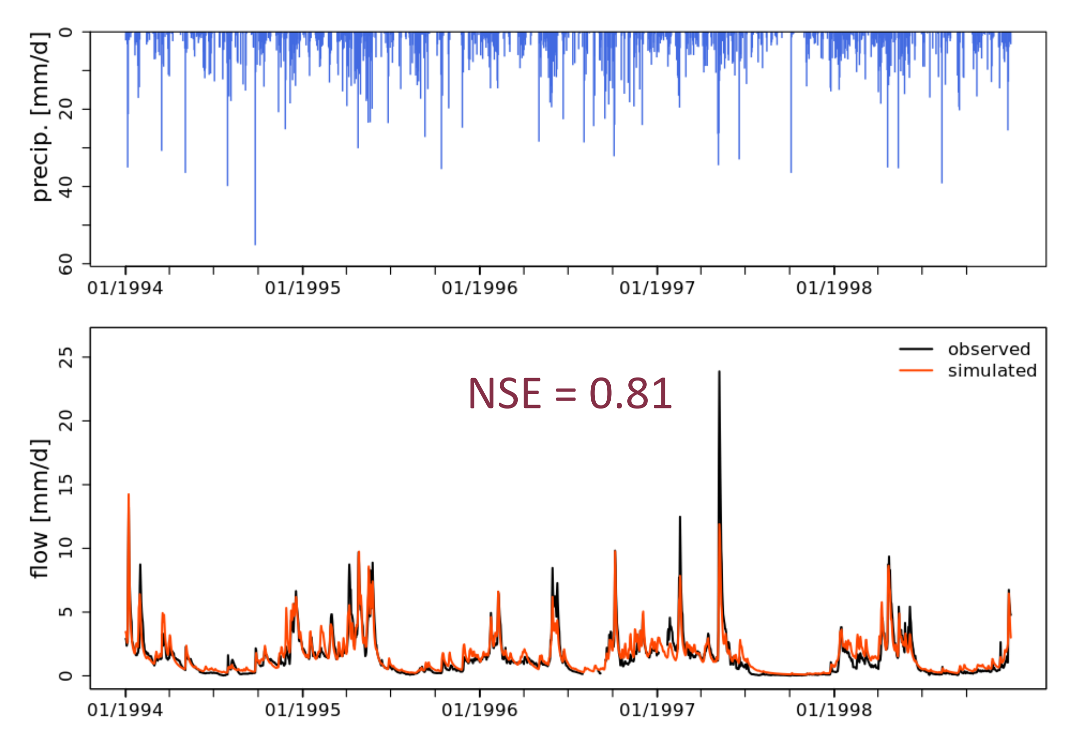
precip. [mm/d]
0
20
40
60
01/1994
01/1995
01/1996
01/1997
01/1998
flow [mm/d]
NSE = 0.81
observed
simulated
0
5
10
15
20
25
01/1994
01/1995
01/1996
01/1997
01/1998

Exercise 3

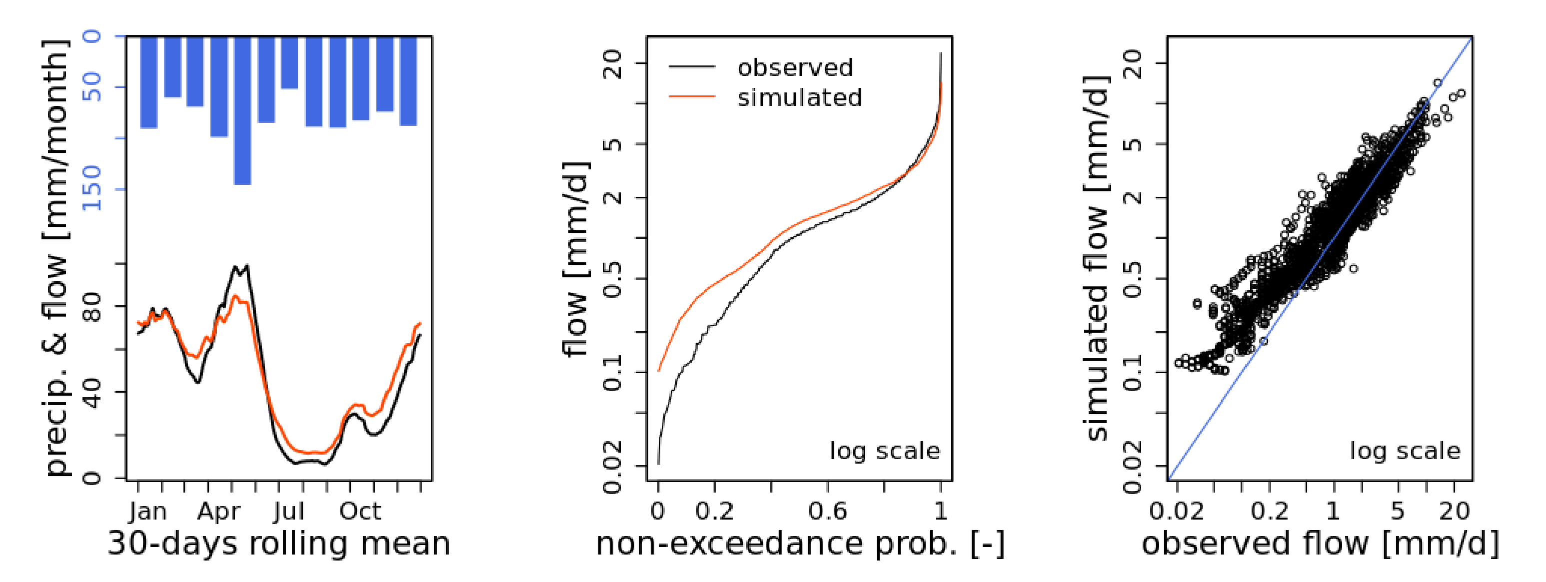


precip. [mm/d]
0
20
40
60
01/1994
01/1995
01/1996
01/1997
01/1998
flow [mm/d]
observed
simulated
NSE = 0.81
25
20
15
10
5
0
flow error [mm/d]
0
-5
-10
Jan 1994
Jan 1995
Jan 1996
Jan 1997
Jan 1998

Exercise 3



precip. [mm/d]
0
20
40
60
01/1994
01/1995
01/1996
01/1997
01/1998
flow [mm/d]
0
5
10
15
20
25
observed
simulated
NSE = 0.81



precip. & flow [mm/month]
0
50
150
0
40
80
Jan
Apr
Jul
Oct
30-days rolling mean
flow [mm/d]
observed
simulated
0.02
0.1
0.5
2
5
20
0
0.2
0.6
1
log scale
non-exceedance prob. [-]
simulated flow [mm/d]
0.02
0.2
1
5
20
log scale
observed flow [mm/d]

4 EVALUATING A MODEL

4.1 Validating a Hydrological Model

Validation versus verification

Objective of validation:

Checking that the model provides an **adequate representation of the hydrosystem**, adapted to the targeted objective.

The objective of verification:

To check that the model is **realistic**, i.e., that it provides a representation that is as close as possible to reality.

Uncertainties - sources

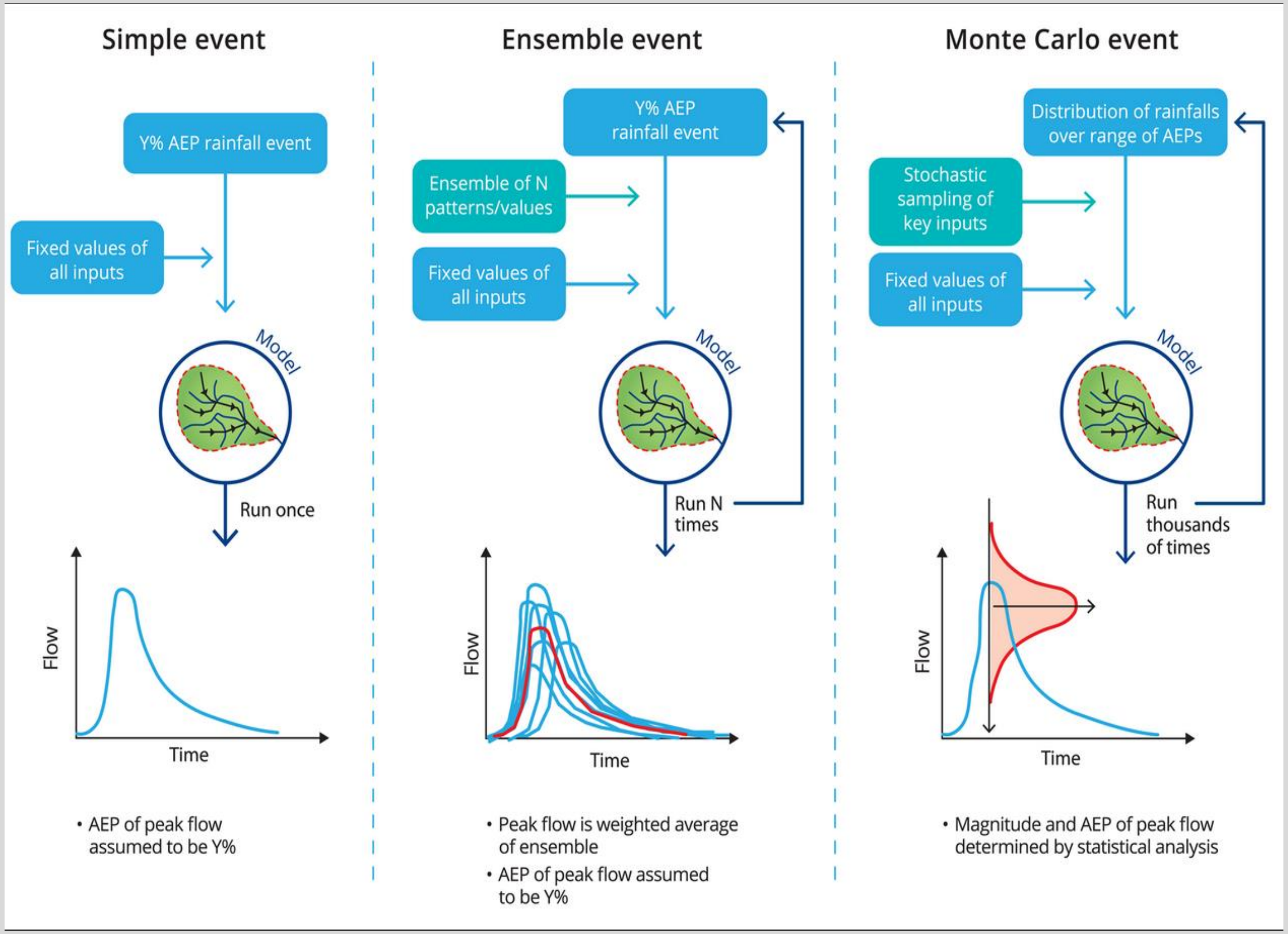
- There are many sources of error and uncertainty. They are mainly related to: i) the **model itself** and ii) the **data** used to calibrate, validate and more generally use the model operationally.
- The uncertainties can be categorised into (1) **measurement**, (2) **structural** and (3) **parameter** uncertainties.

Uncertainties - analysis

Analysing uncertainties

- **Sensitivity analysis** of the effect of input data and model parameters on model output
- **Model structure uncertainty analysis**: 1) calibrating different types of models or different versions of the given model over a given period and 2) comparing results of their simulations in another context

Determination of the most probable scenario



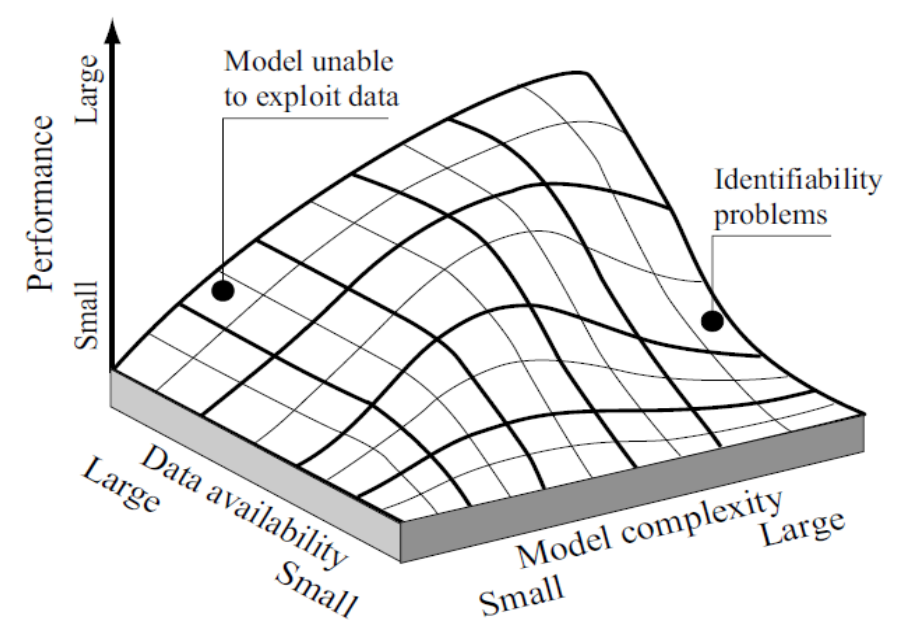
(Ball et al., 2016)

5 CHOOSING AND USING A HYDROLOGICAL MODEL

- Data Availability, Model Complexity and Performance

- Model Output Requirements

- Choosing a Model Type



Grayson and Blöschl, 2000

Examples of hydrological models

- Lumped Conceptual Model—the **GR4J** Model

- Semi-Distributed Conceptual Model—the **HBV** Model

- Distributed Physically-based Conceptual Model—**TOPMODEL**, the basis for **TopNet**, the NZ nationwide hydrological model developed and operated by ESNZ (formerly NIWA)

- Distributed Physically-based Hydrological Model: The **MIKE SHE** Model

- Semi-distributed **HEC-HMS** (Hydrologic Engineering Center-Hydrologic Modelling System)

Useful things to consider when working on a project that involves modelling

- **Evaluate/check data** and reference data sources (quality control – avoid ‘rubbish in rubbish out’)
- **Document** all **assumptions** (you will forget)
- **Plan and control computer runs** (this may become costly)
- **Review** reasonableness of **output** (your expertise will allow you to judge the results – do the results make sense?)
- **Document your results** (the good, the bad, the ugly,…all will be easier the second time around)

Selected software and data sources

Software

- HEC-HMS – installed on many computer lab computers, available on UC Software Centre and freely available online

  http://www.hec.usace.army.mil/software/hec-hms/

Data

- ESNZ High Intensity Rainfall System:

http://hirds.niwa.co.nz/

- ESNZ DataHub, includes National Climate Database - ESNZ CliFlo

https://data.niwa.co.nz/

- River flow data:

    Regional and city councils, for example Environment Canterbury
    https://www.ecan.govt.nz/data/riverflow/

- ESNZ hydro web portal:

https://niwa.co.nz/niwa-hydro-web-portal

Sample questions

Check your understanding

1. Explain the principles and objectives of hydrological modelling.
2. Explain the structure and variables of a hydrological model.
3. Explain the main types of hydrological models.
4. Explain the steps to develop a hydrological model.
5. Develop a simple hydrological model by selecting the required components (desktop exercise).
6. Explain the challenges of model calibration and validation.
7. Discuss the importance of considering uncertainties and judge the level of relevance of various uncertainty sources for a given application.

Check your understanding

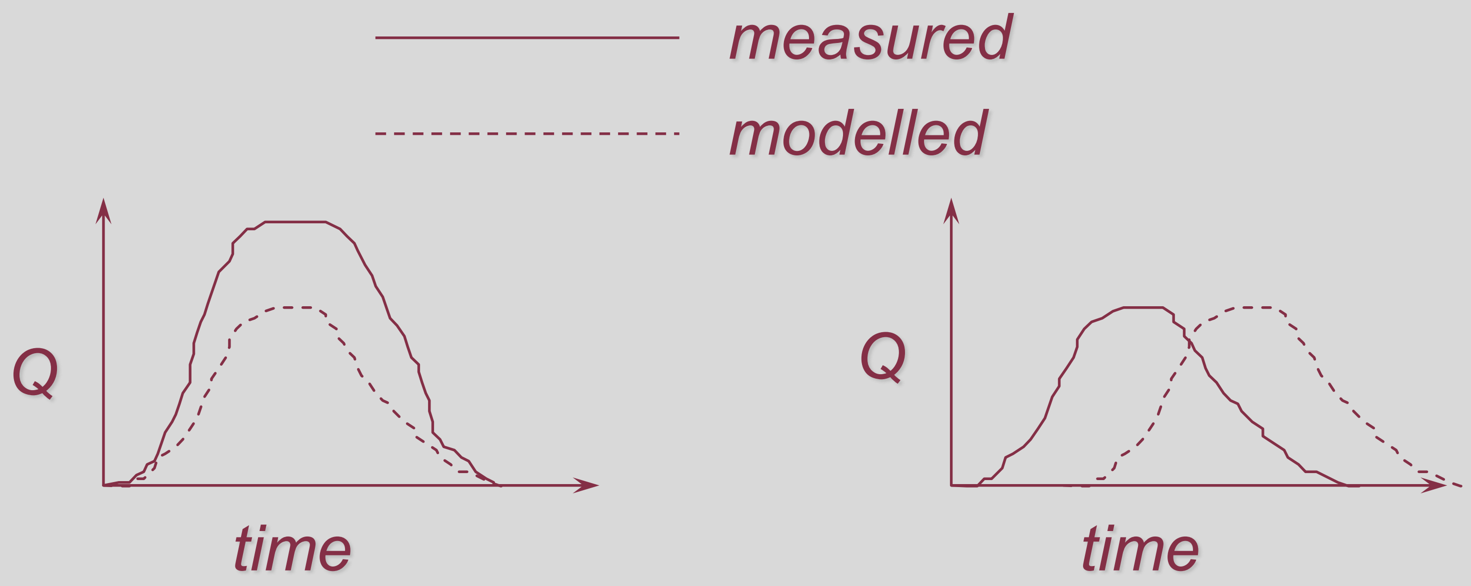
What are

- State variables
- Parameters
- Boundary conditions
- Initial conditions

Check your understanding Calibration

A hydrological model has been developed. Routing is modelled using the Muskingum method. The method has two parameters: X and K

Which parameter would you adjust to improve the simulation results shown on the left and on the right in the Figure below?

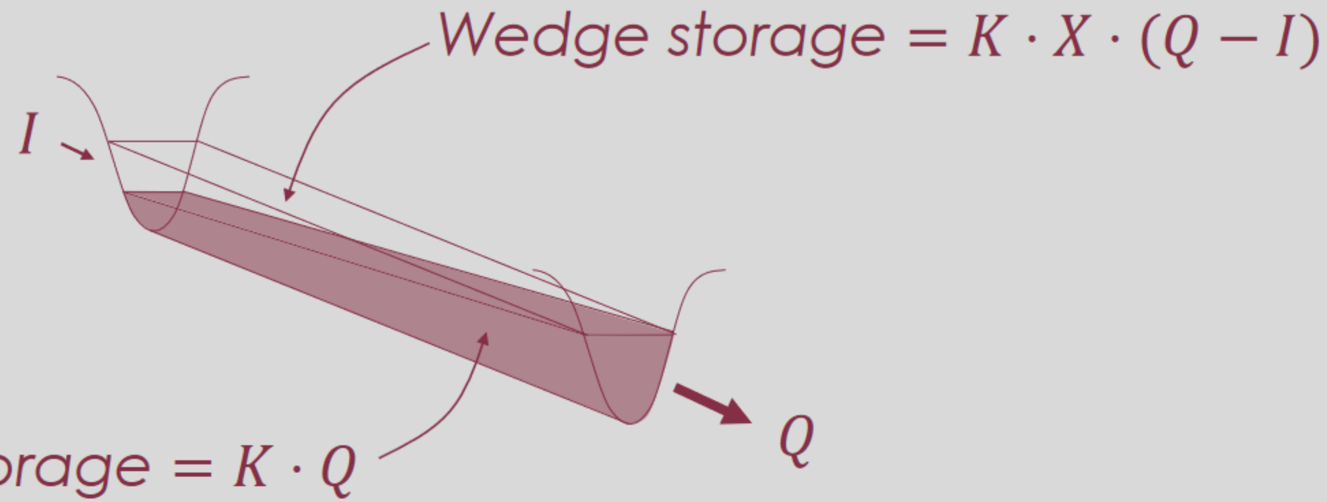


Check your understanding
Calibration

From ENCN342:

Flood recession

$I < Q$



$$S = K[X \cdot I(t) + (1 - X) \cdot Q(t)]$$

Check your understanding

Given your project must consider a river with a short record of historical flow measurements, but a long time series of daily precipitation measurements in the river catchment.

The Regional Council would like to know from you: What is the amount of water that can be withdrawn sustainably from the river to satisfy water demand (QD) for irrigation and water supply?

How will you approach this task?